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ENVIRONMENTAL PUBLIC HEALTH TRACKING PROGRAM

Geographic Aggregation Tool

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Geographic Aggregation

Merging small geographic areas with neighboring areas. Alternative to smoothing.

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Key Objectives & Benefits

- **Improve Statistical Reliability:** Mitigates the issue of unstable or unreliable rates caused by small sample sizes and random fluctuations.
- **Minimize Data Suppression:** Reduces the need to completely suppress data, allowing for more comprehensive and meaningful public health reporting.
- **Protect Confidentiality:** Prevents the disclosure of sensitive or personally identifiable health information in sparsely populated areas.
- **Ensure Regulatory Compliance:** Aligns data practices with state laws, agency policies, and federal grant requirements.
  - Environmental Public Health Tracking (EPHT) Grant guidelines.

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Health data can be shown at different geographic scales

- Residential address
- Census blocks, and tracts
- Towns
- Counties
- State

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Concerns about release of small area data

- Risk of disclosure of confidential information.

- Rates of disease are unreliable due to small numbers.

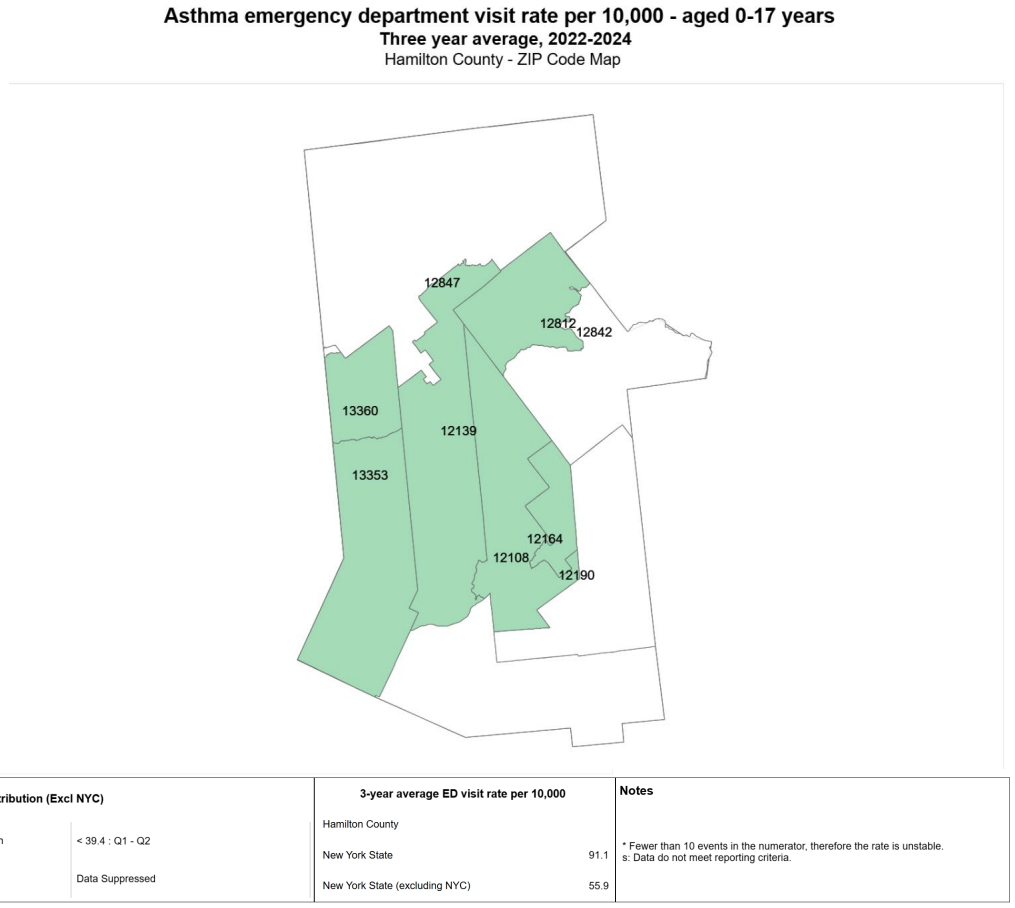
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Rate maps with small numbers provide very little information.

- Rates are suppressed due to confidentiality or unstable.

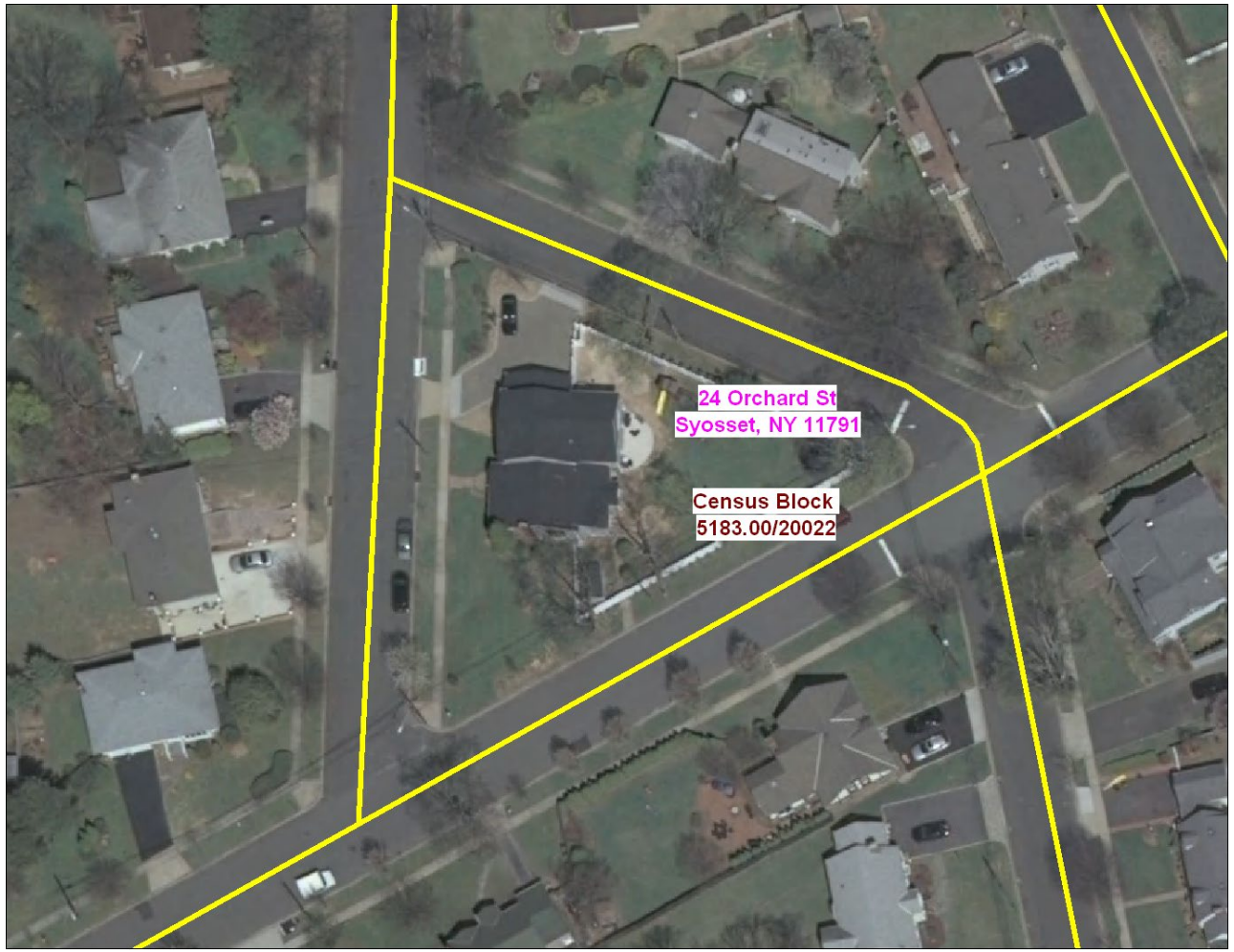
Asthma emergency department visits
0-17 years, 2022-2024

ZIP Code	ED visits (3-year total)	3-year average ED visit rate per 10,000
12108	0	0.0*
12139	0	0.0*
12164	0	0.0*
12812	0	0.0*
13353	0	0.0*
13360	0	0.0*
12190	s	s
12842	s	s
12847	s	s



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Disclosure of confidential information



Census Blocks

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Smoothed or Aggregated Count & Rate Maps

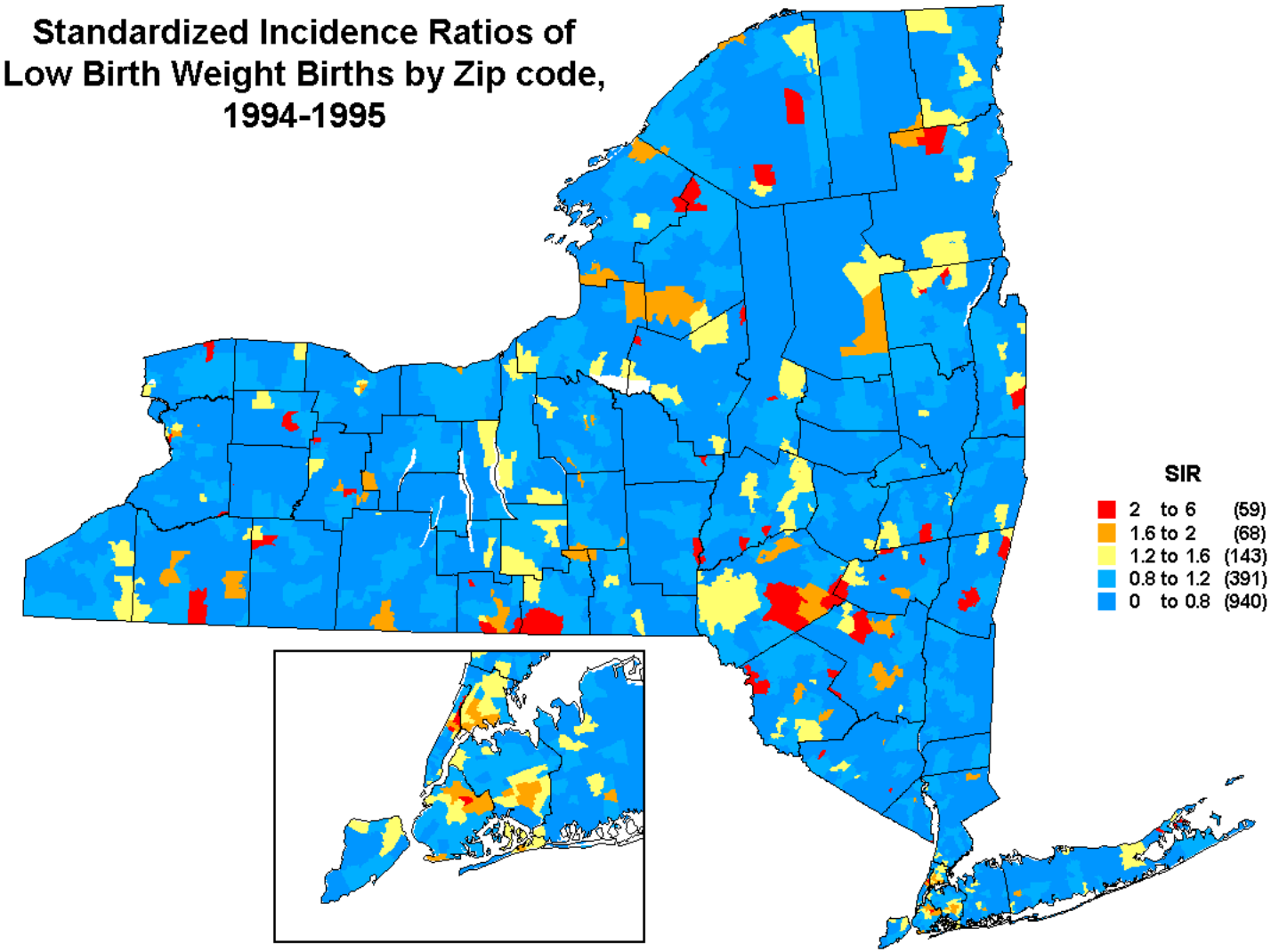
- Protect Confidentiality so data can be shared.

- Reduce random fluctuations in rates due to small numbers.

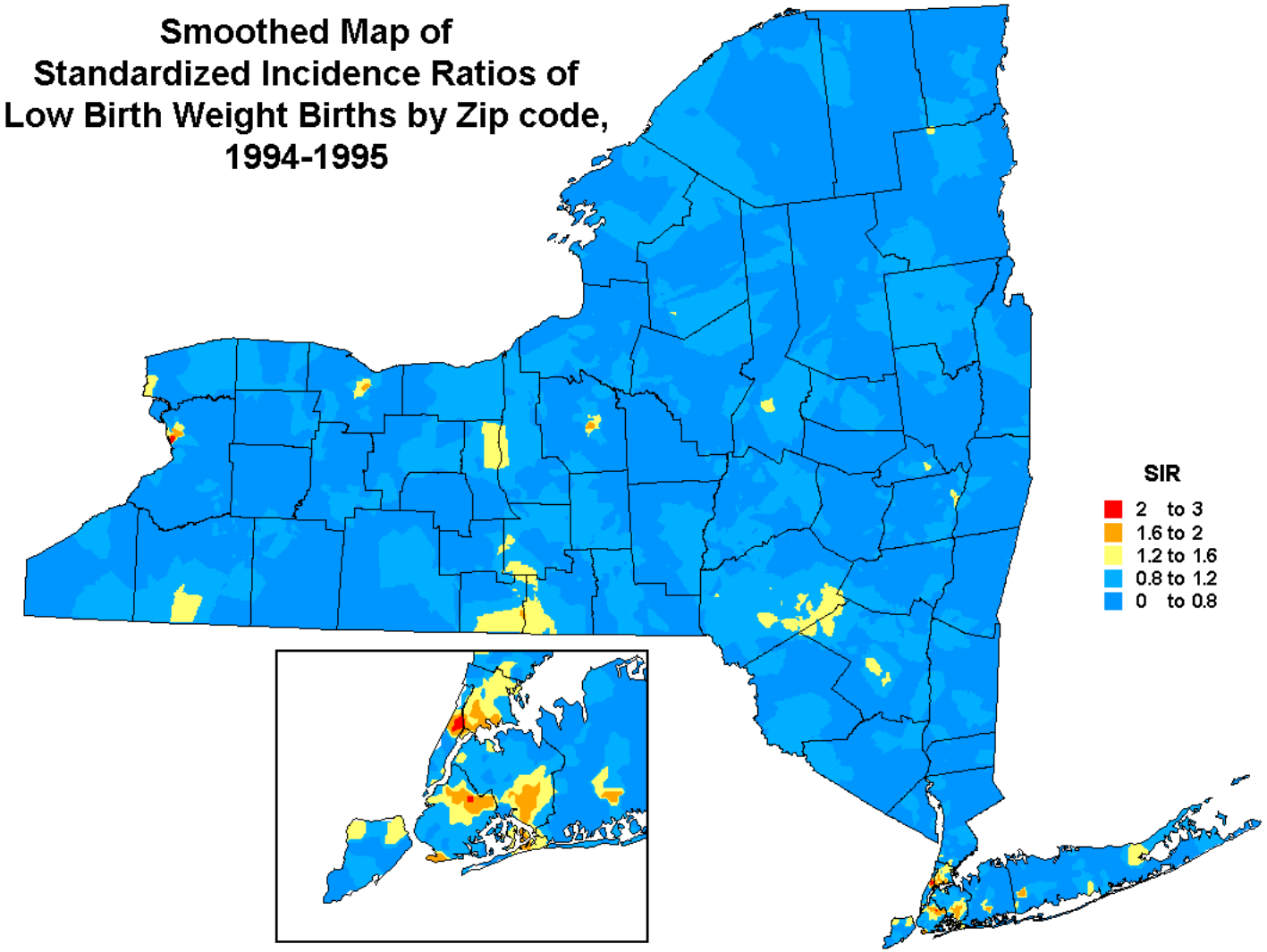
Smoothed Rate Maps

- Borrow data from neighboring areas to provide more stable rates of disease.
  - Shareware tools available
  - Empirical or Hierarchal Bayesian approaches
  - Adaptive Spatial Filters
  - Head banging
  - etc.

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Based on grid points capturing a minimum of 250 births

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Problems with smoothing

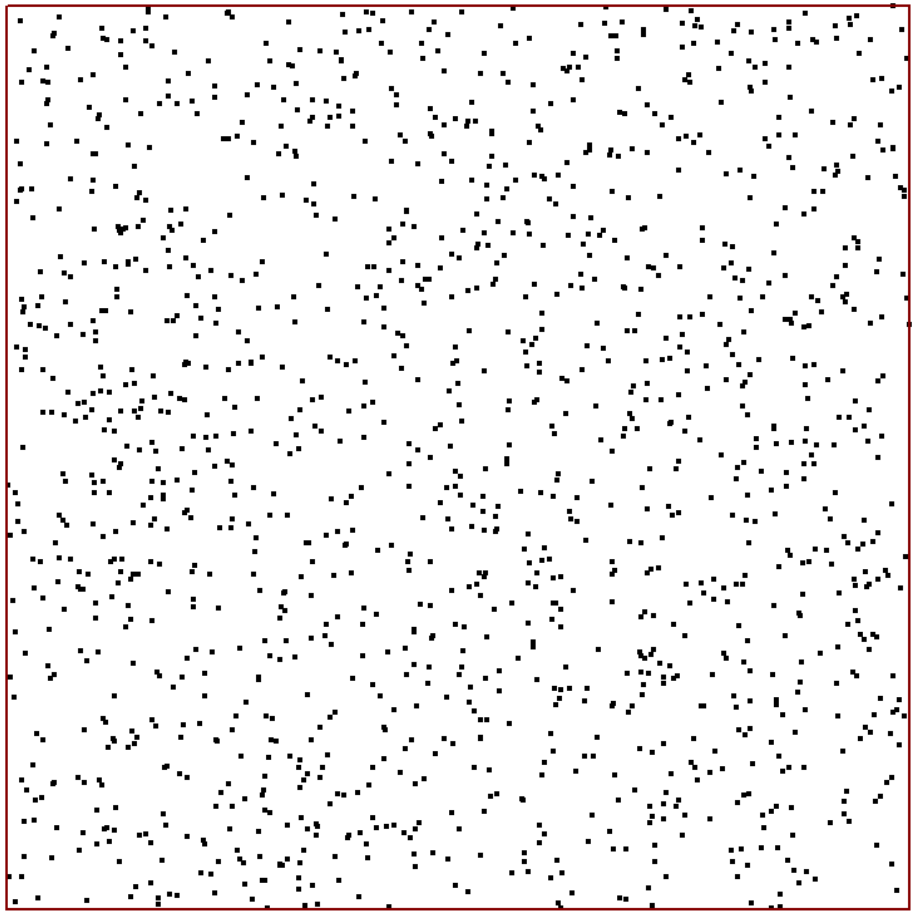
- Does not provide counts & rates for defined geographic areas.
- Not clear how to link risk factor data with smoothed health data.
- Methods are sometimes difficult to understand - “black boxes”
- Does not meet requirements of some recent New York policies & legislation.

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Modifiable Area Unit Problem (MAUP)

- Change in scale of areal units cause changes in rates of disease.
- Changes to the shape or location of the zones can dramatically alter where rates are high and low.
- Take the following example where random points are placed on a map, and the zones are slightly moved.
- The expected number of points in each zone is 10
- Notice how the number of points change in the zones even though the scale and placement of the points remain the same.

Modifiable Area Unit Problem (MAUP)

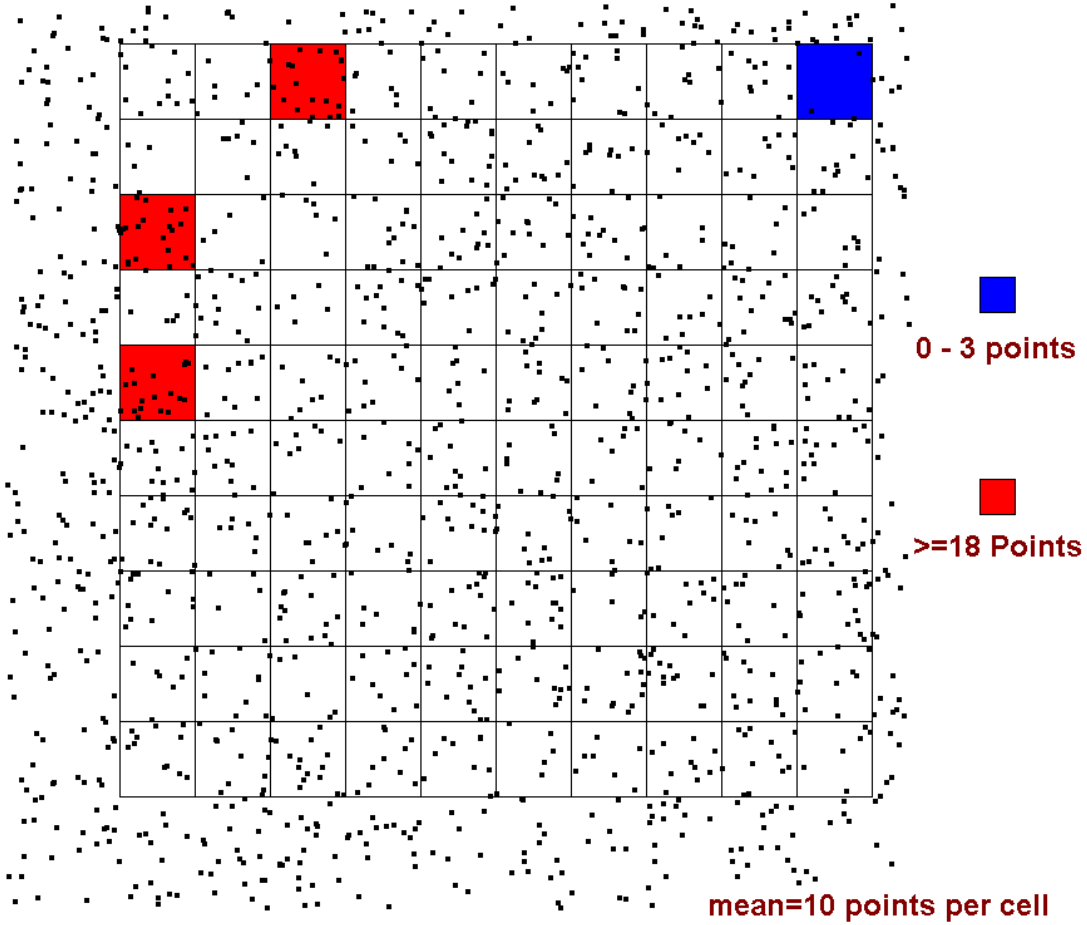


Random Distribution of points

Talbot

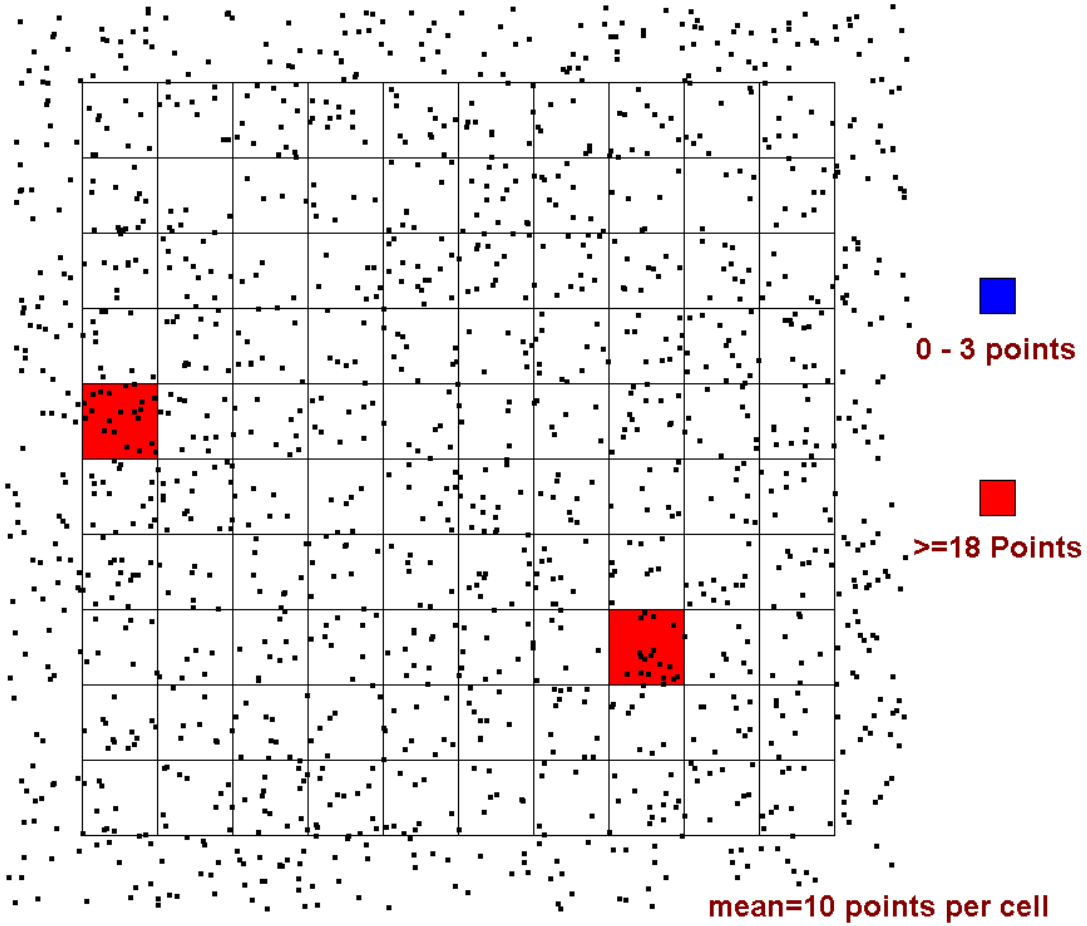
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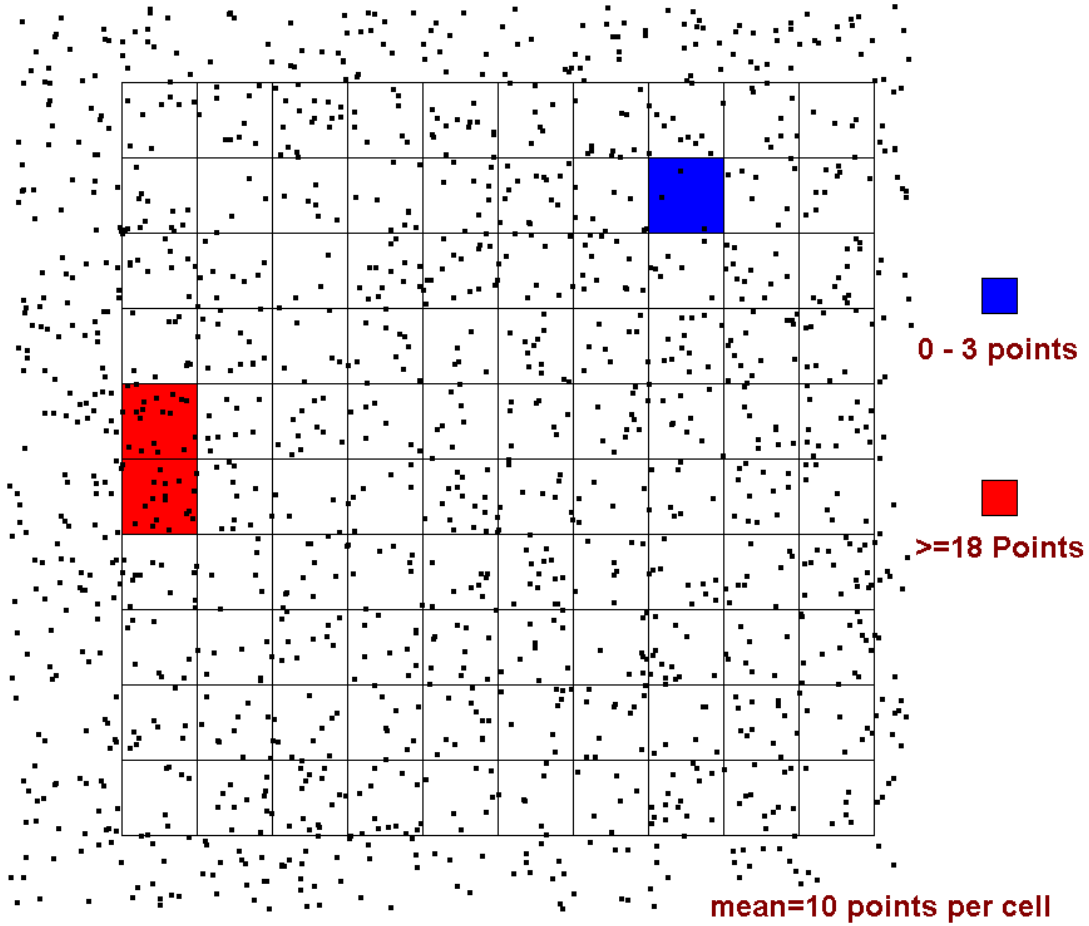
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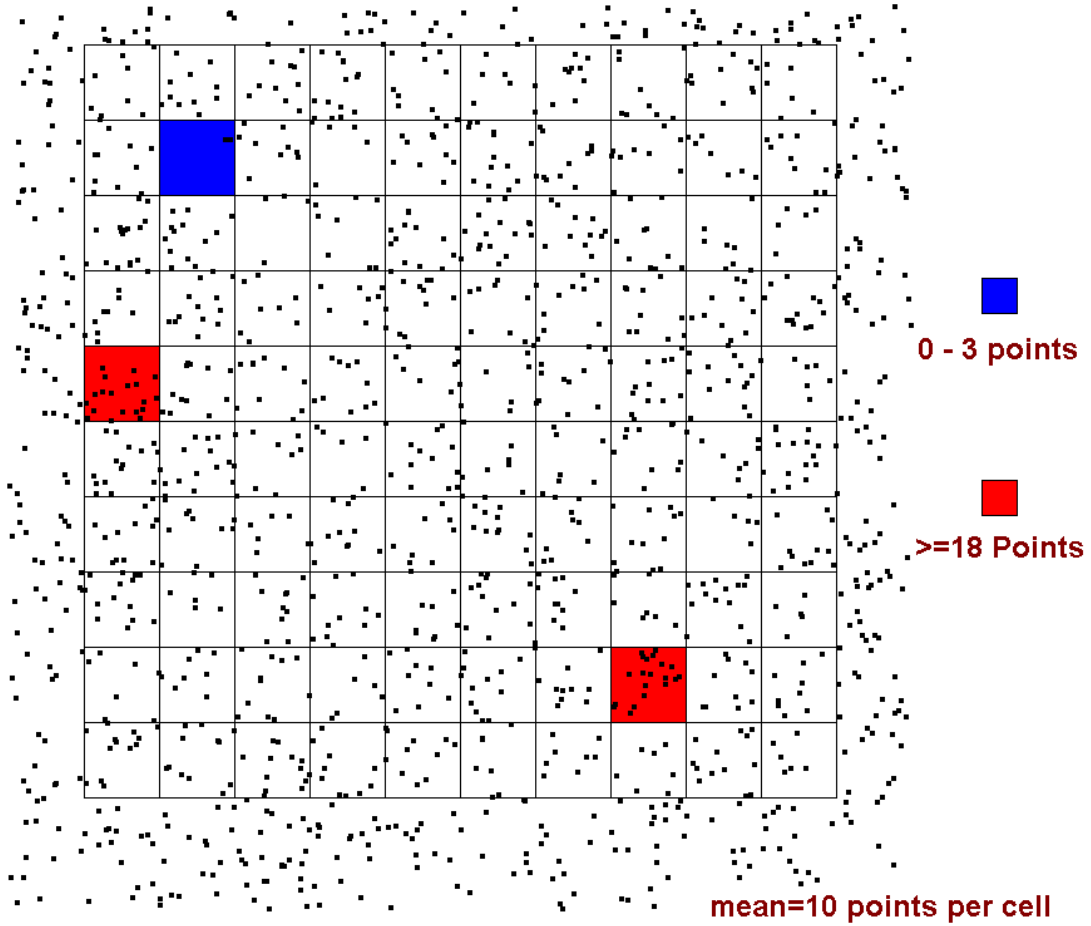
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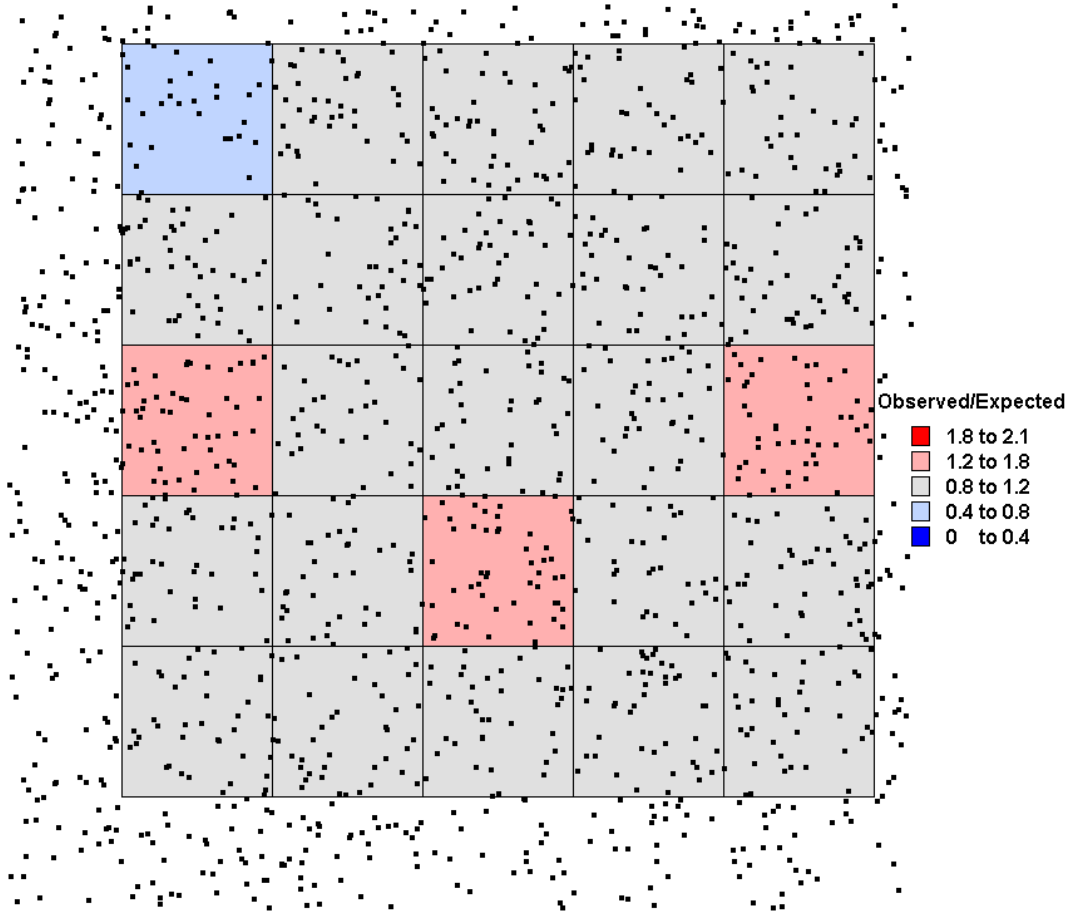
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Modifiable Area Unit Problem (MAUP)



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Modifiable Area Unit Problem (MAUP)



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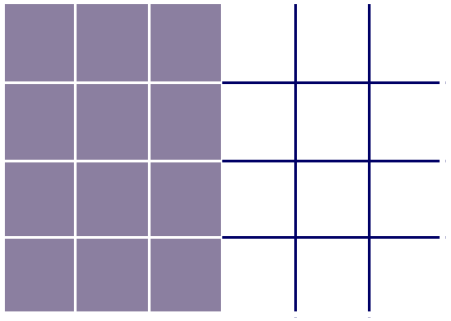
Aggregated Count or Rate Maps

- Merge small areas with neighboring areas to provide more stable rates of disease and/or protect confidentiality.
  - Manual aggregation may be time consuming
  - Existing automated tools use expensive proprietary software
  - Existing automated tools not geared toward public health needs.

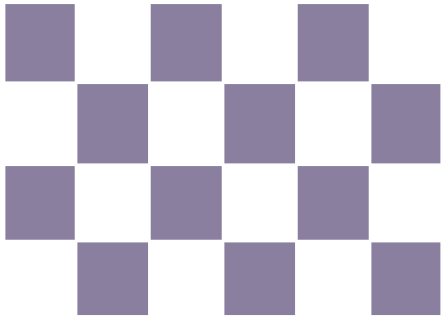
Spatial Autocorrelation

"Everything is related to everything else, but near things are more related than distant things."

- *Tobler's first law of geography*



Positive autocorrelation

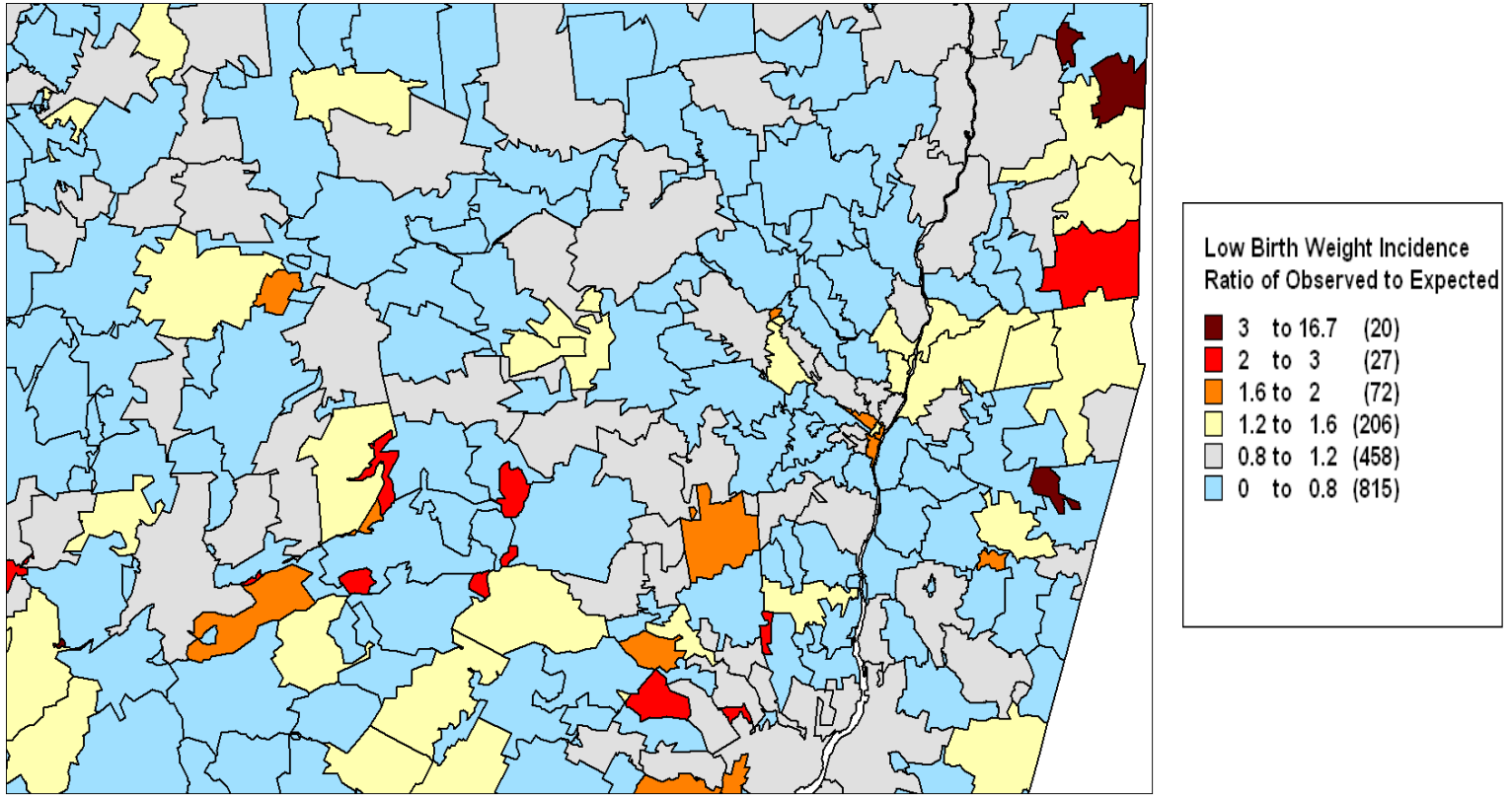


Negative autocorrelation

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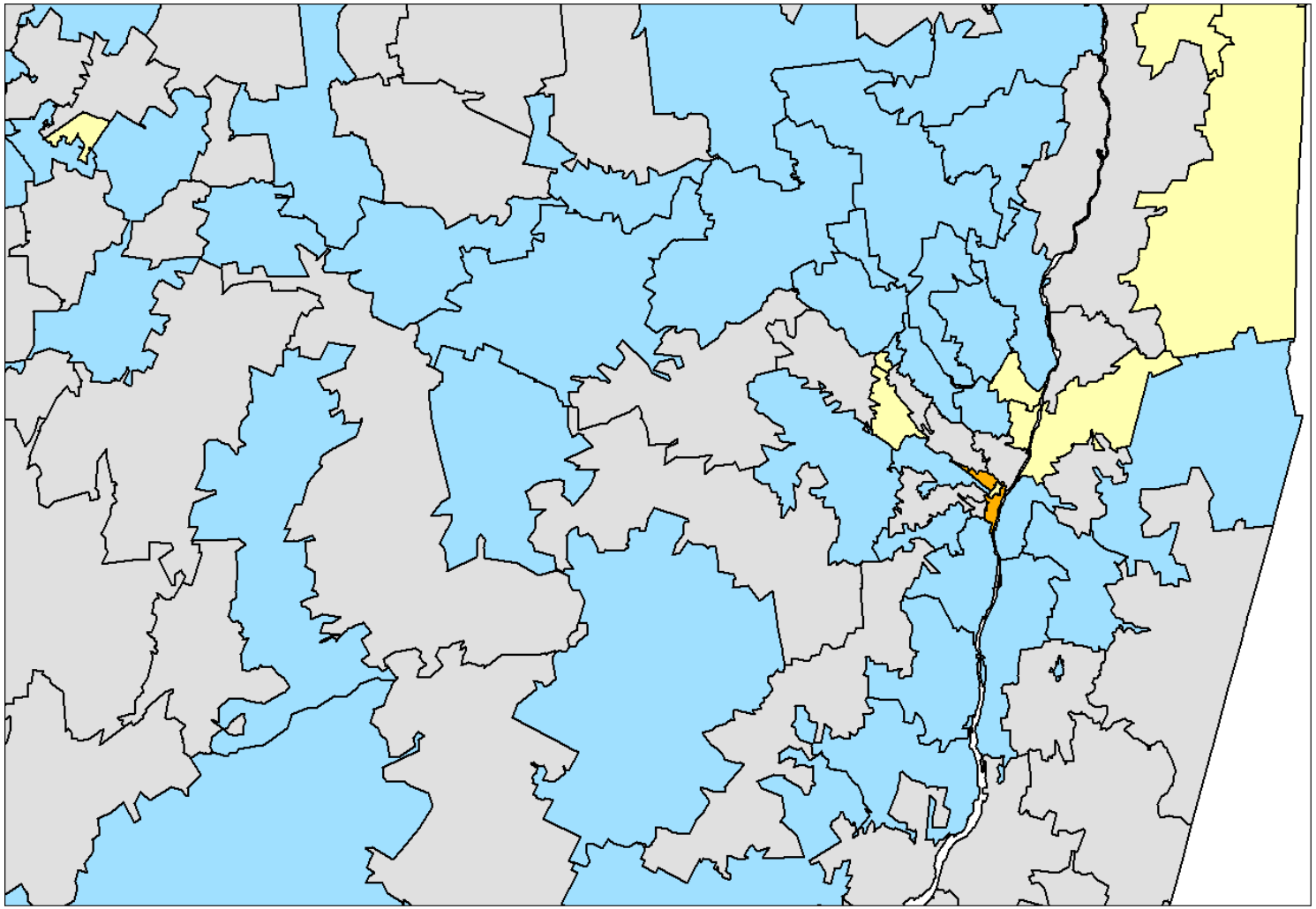
Original ZIP Codes

3 Years Low Birth Weight Incidence Ratios



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Aggregated to 250 Births per ZIP Code Group



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Geographic Aggregation Tool: Advantages

- Aggregate small areas into larger ones.
- User decides how much aggregation is needed.
  - Based on cases and/or underlying population
  - Examples: minimum population of 7,500 or 15,000
- Works with various levels of geography.
  - Census blocks, tracts, towns, ZIP codes etc.
  - Can nest one level of geography in another
- Uses open source free software.
- Can use the results in mapping programs.

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Geographic Aggregation Tool Requirements

- R 4.5 or later installed with GAT and other necessary packages
- Area Shapefile
  - Unique identifier for each area
  - Number for each area (ex. Population, # cases)

R Packages

Required:

- classInt
- graphics
- grDevices
- methods
- raster
- RColorBrewer
- sf
- sfdep
- spdep
- stats
- tcltk
- tibble

Optional:

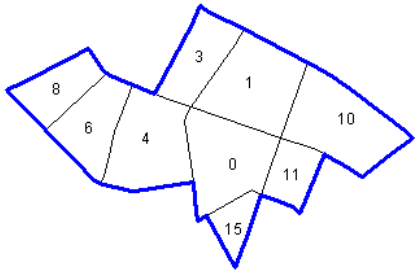
- knitr
- lwgeom
- prettymapr
- rmarkdown
- tools
- RoxygenNote

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Geographic Aggregation Tool

Original Block Data †

Block	Cases
122300/2004	10
122300/2005	11
014500/3005	3
014500/3007	4
014500/3008	0
014500/3009	1
014500/3010	15
103202/2001	8
103202/2002	6

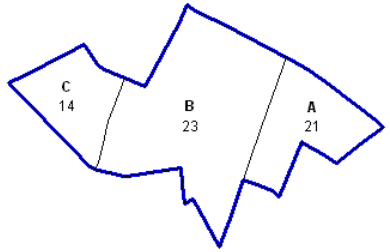


R Tool

Regions

Block	Cases	Region
122300/2004	10	A
122300/2005	11	A
014500/3005	3	B
014500/3007	4	B
014500/3008	0	B
014500/3009	1	B
014500/3010	15	B
103202/2001	8	C
103202/2002	6	C

Cases	Region
21	A
23	B
14	C

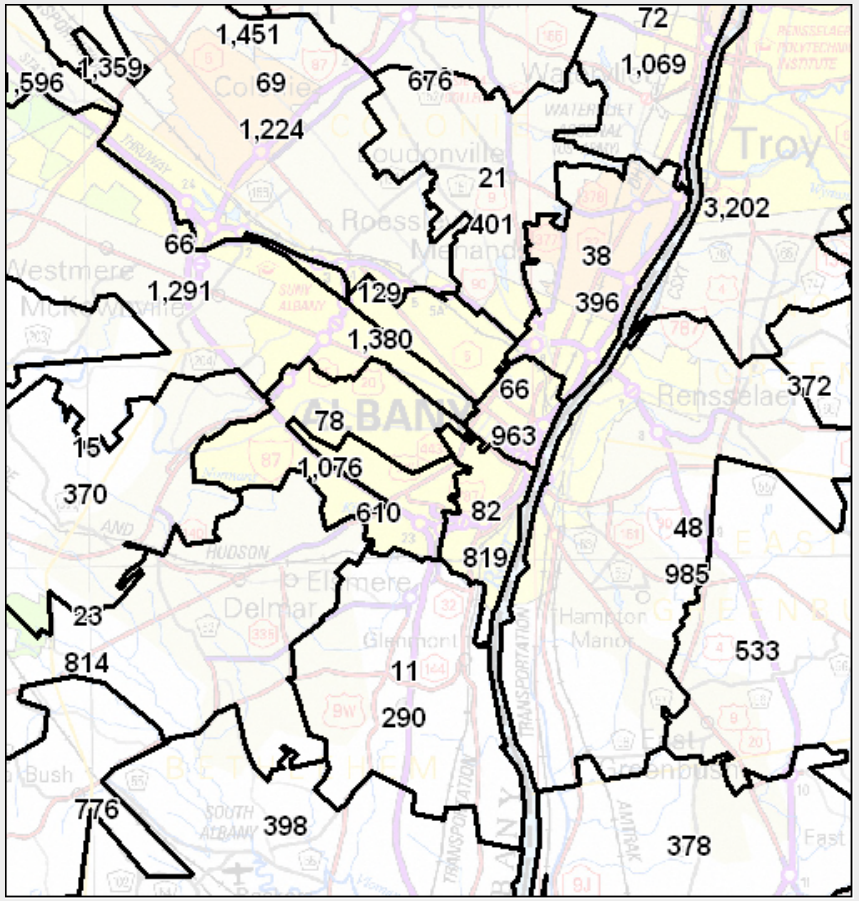


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† Simulated data

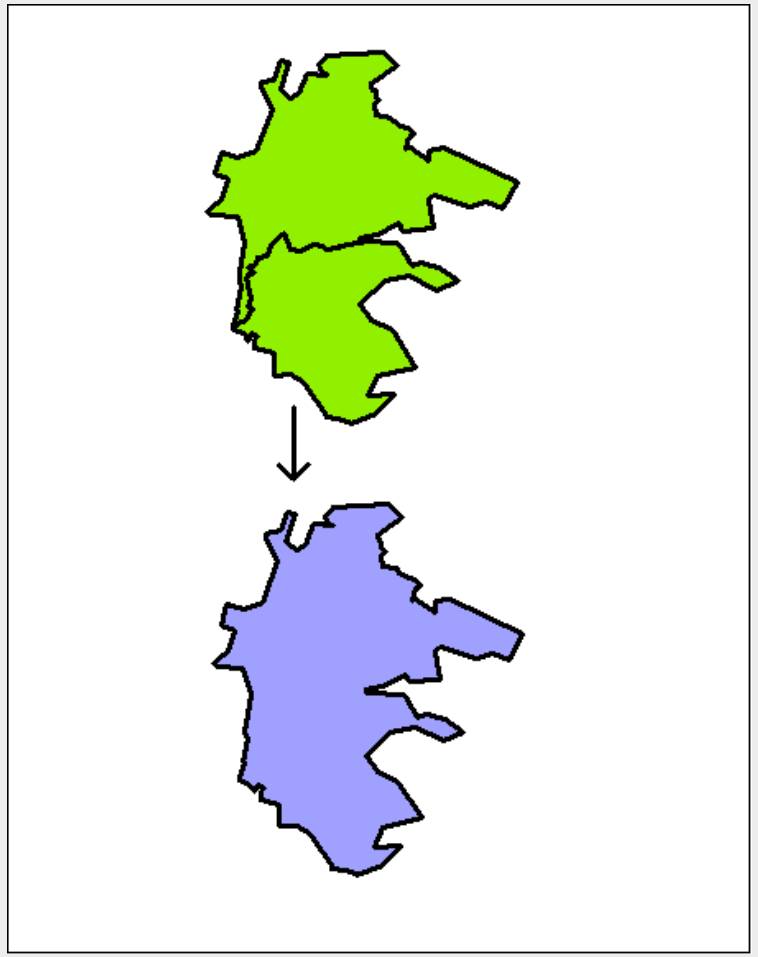
The Aggregation Process: goals

- Still have many areas after aggregation
- The areas should be reasonably compact
- The areas should have minimum and/or maximum values for population or cases



The Aggregation Process: method

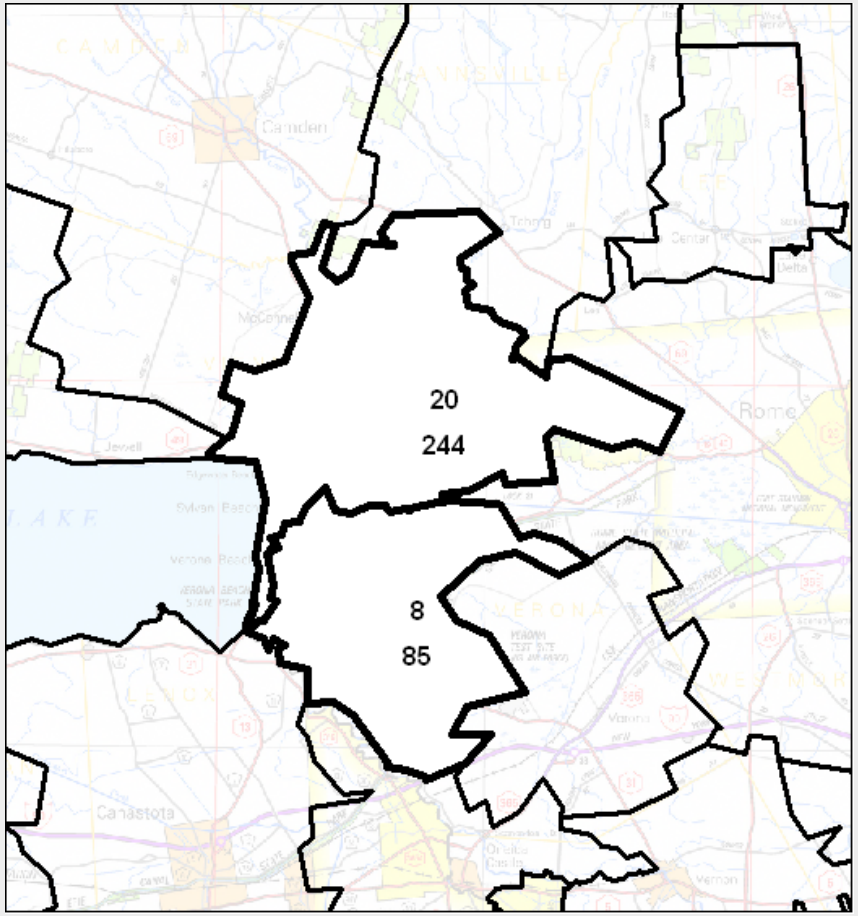
- Pair-wise merges
- Merge until the areas should have minimum values for certain criteria



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The Pairwise Merge: 1st area

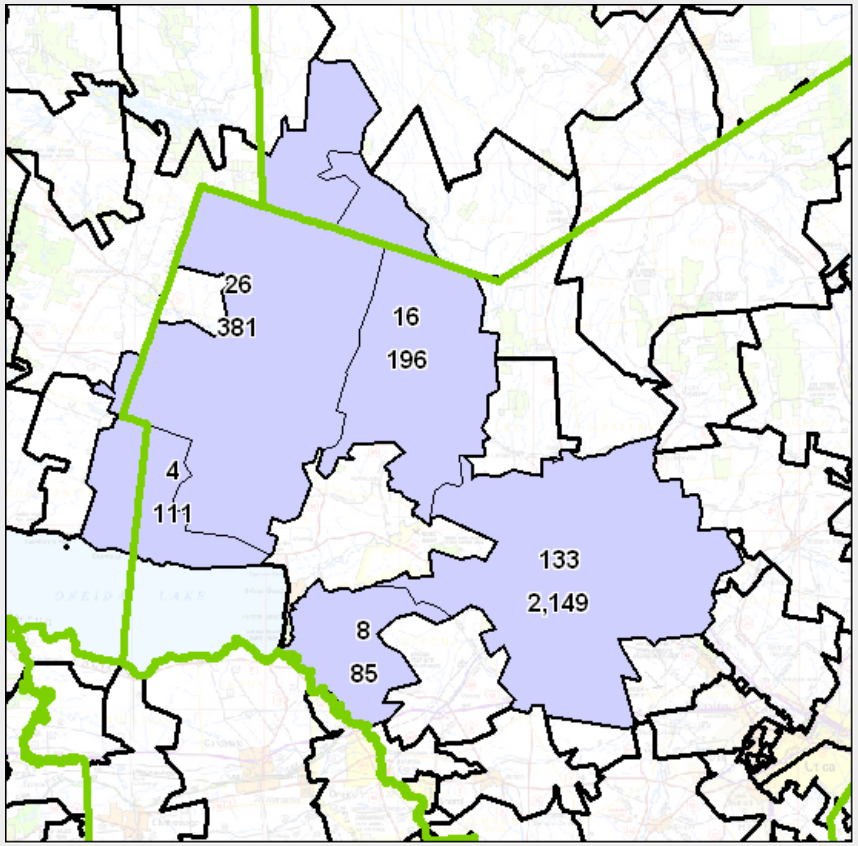
- Select those areas which require merging to meet minimum values
- How to pick a neighbor to merge to?
  - Closest area
  - Lowest number
  - Similarity



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The Pairwise Merge: 2nd area

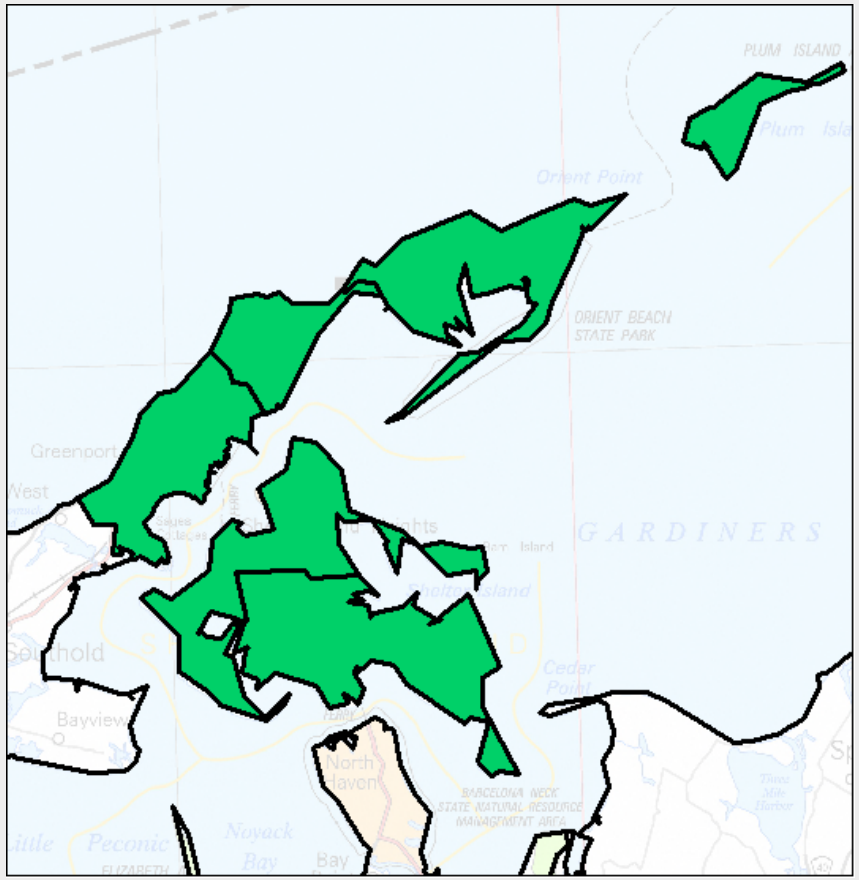
- Find the adjacent neighbors of the selected area
- If a boundary variable is used, select those neighbors that are within the boundary variable



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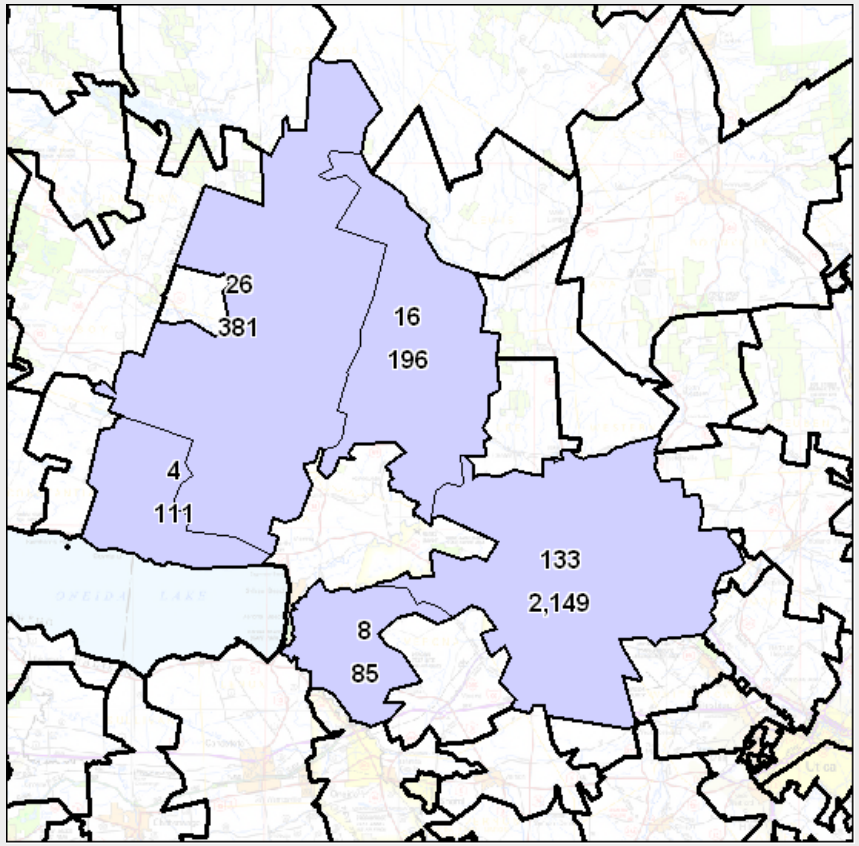
The Pairwise Merge: 2nd area

- If there are no adjacent neighbors, choose the closest area (according to distance between centroids)



Pairwise Merge: three methods to choose 2nd area

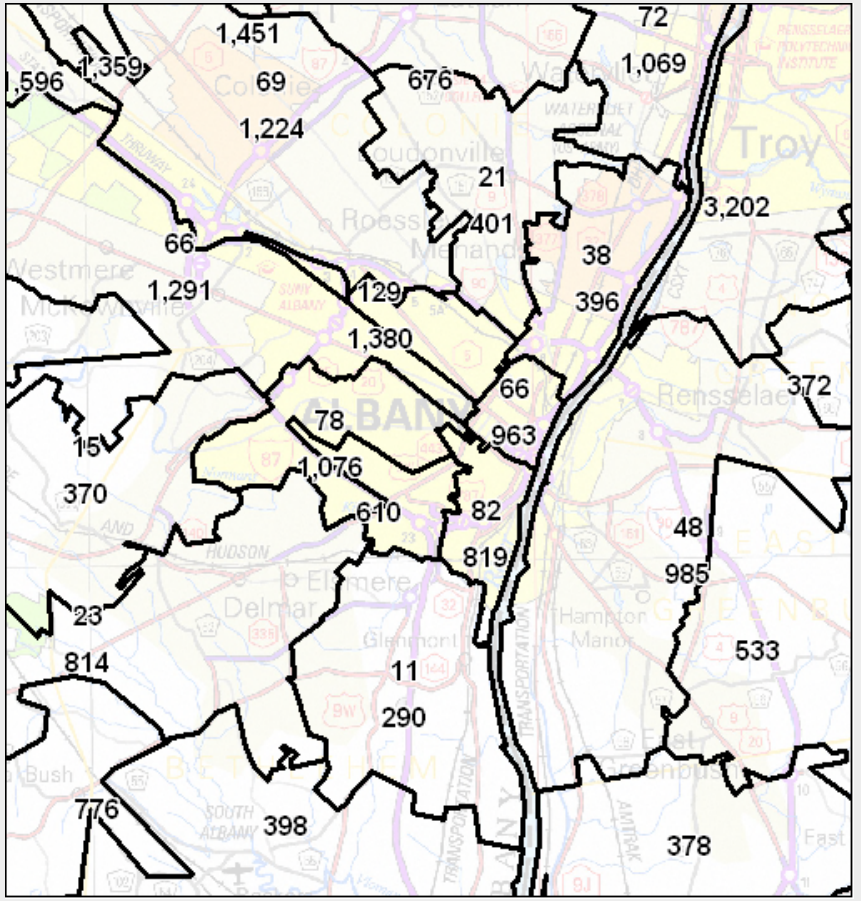
- Closest centroid
  - population-weighted
  - geographic
- Lowest aggregation variable
- Most similar ratio



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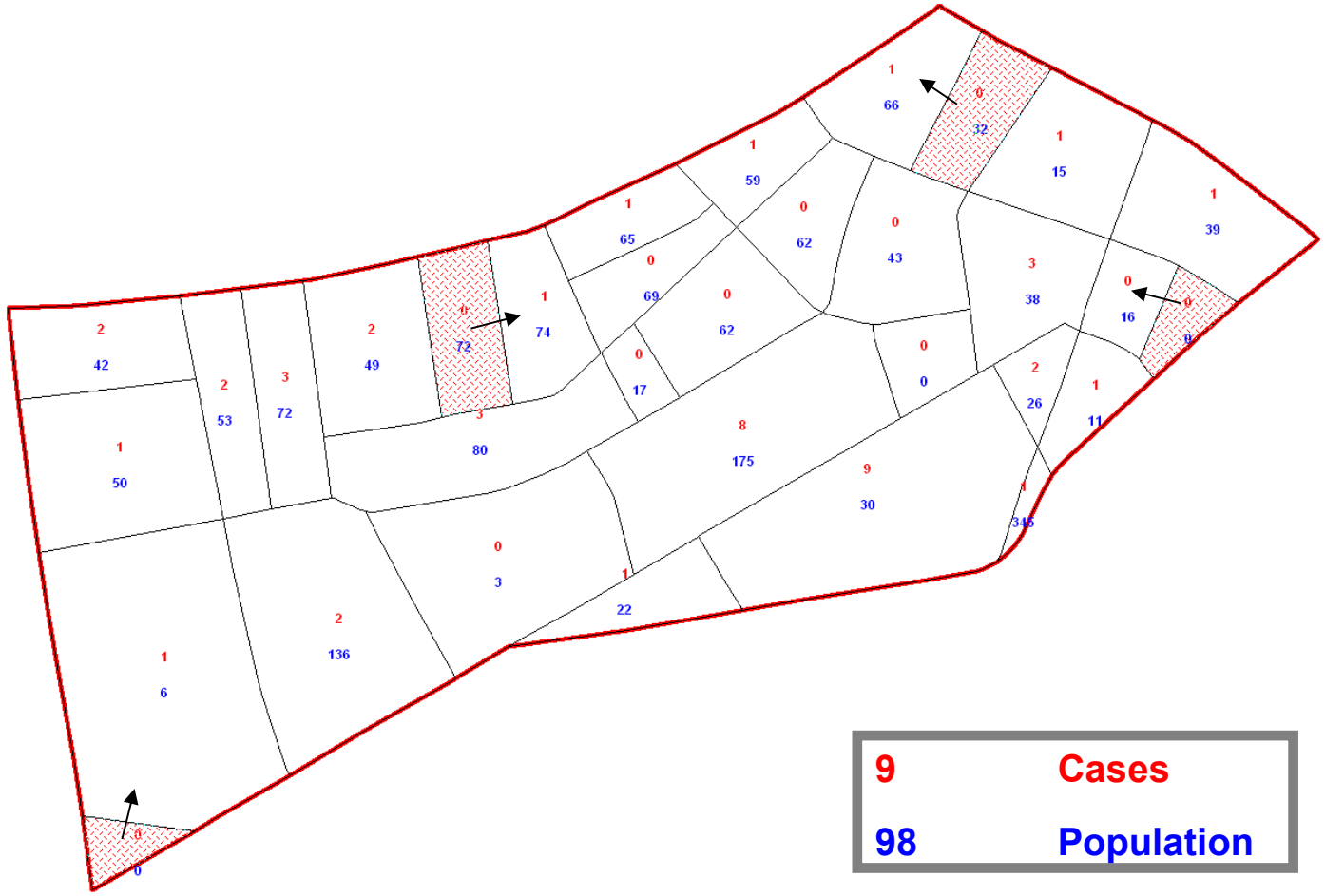
The Aggregation Process: method

- Repeat pair-wise merges
- Merge until the areas should have minimum values for certain criteria



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The Aggregation Process: Example

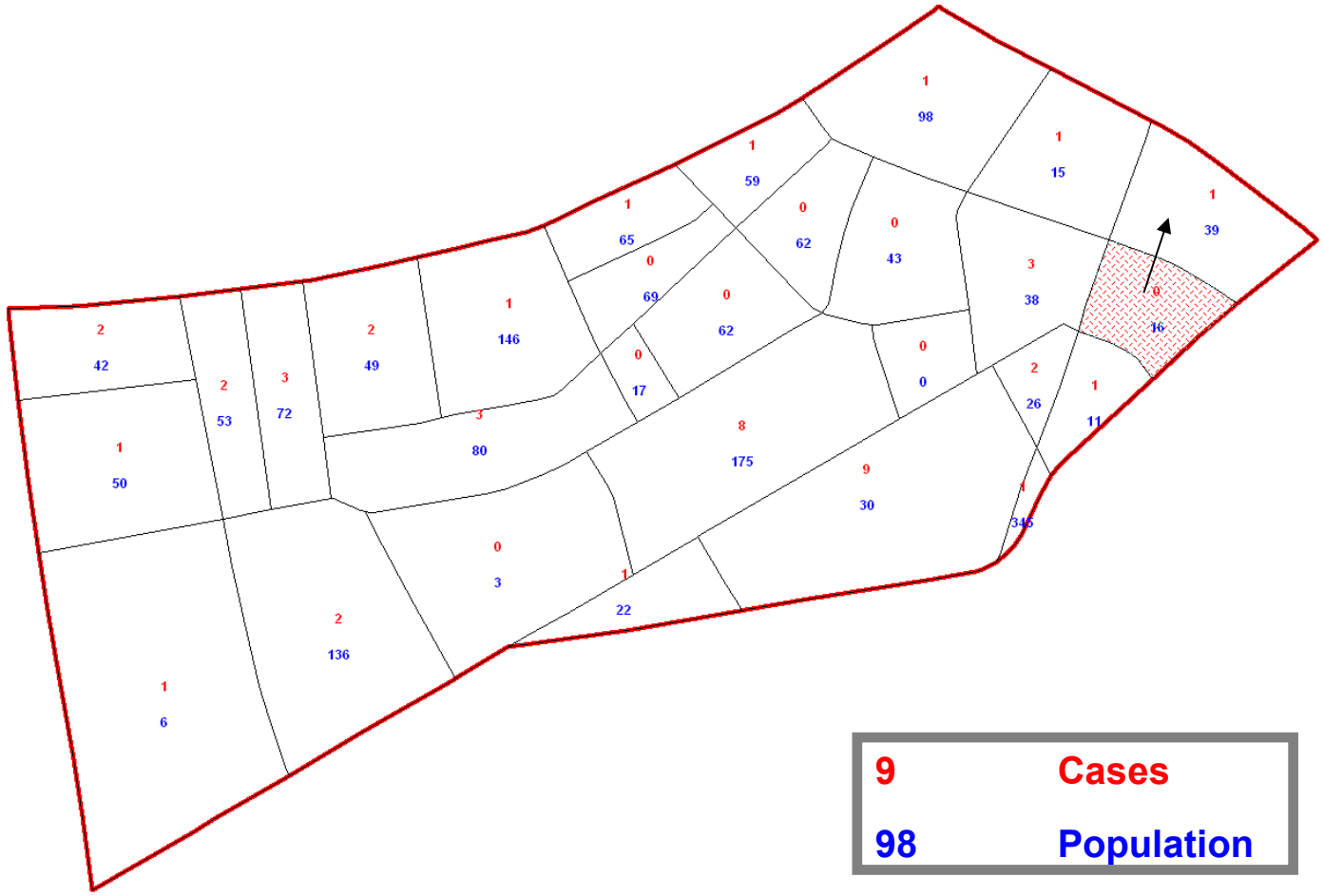


9	Cases
98	Population

† Simulated data

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The Aggregation Process: Example

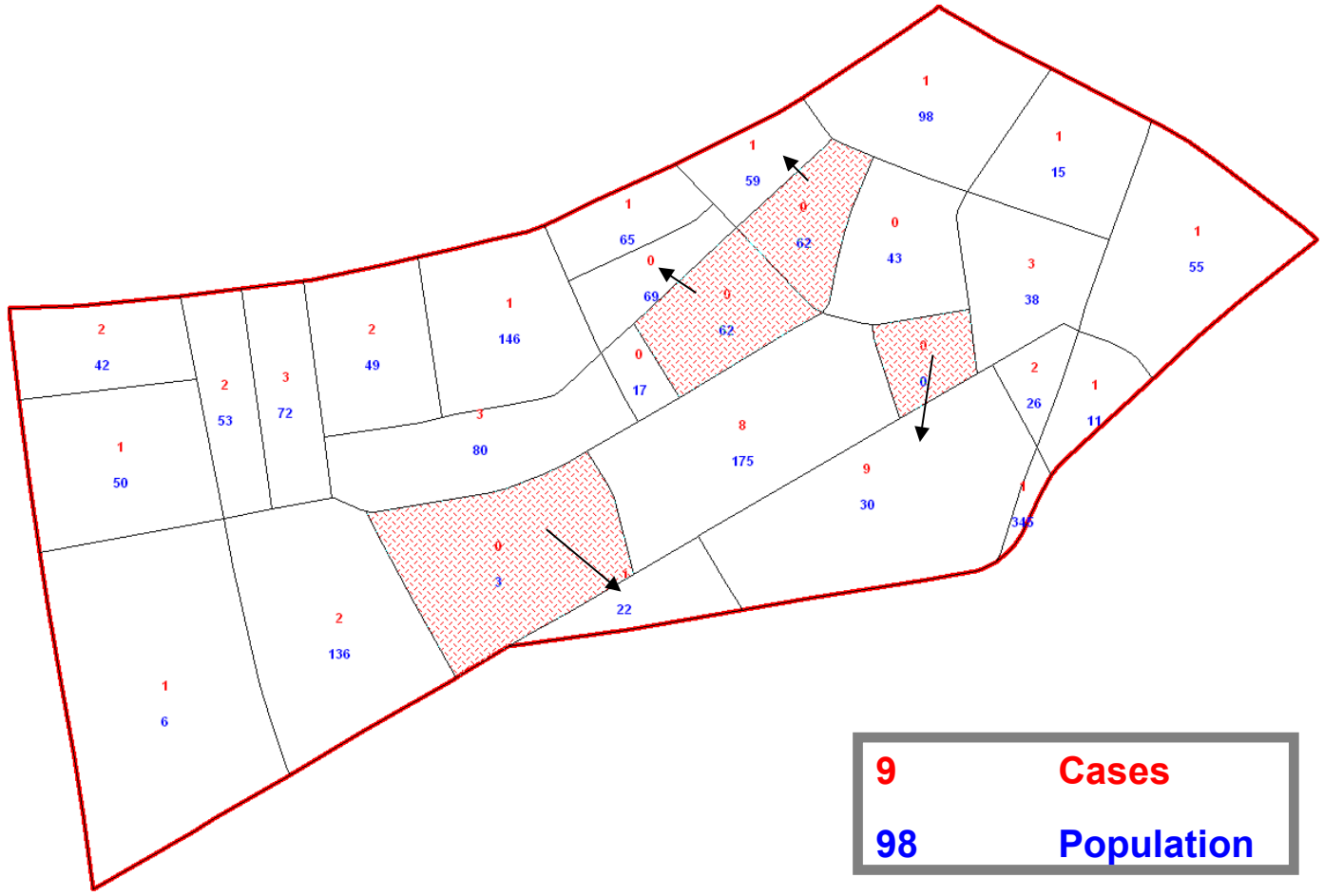


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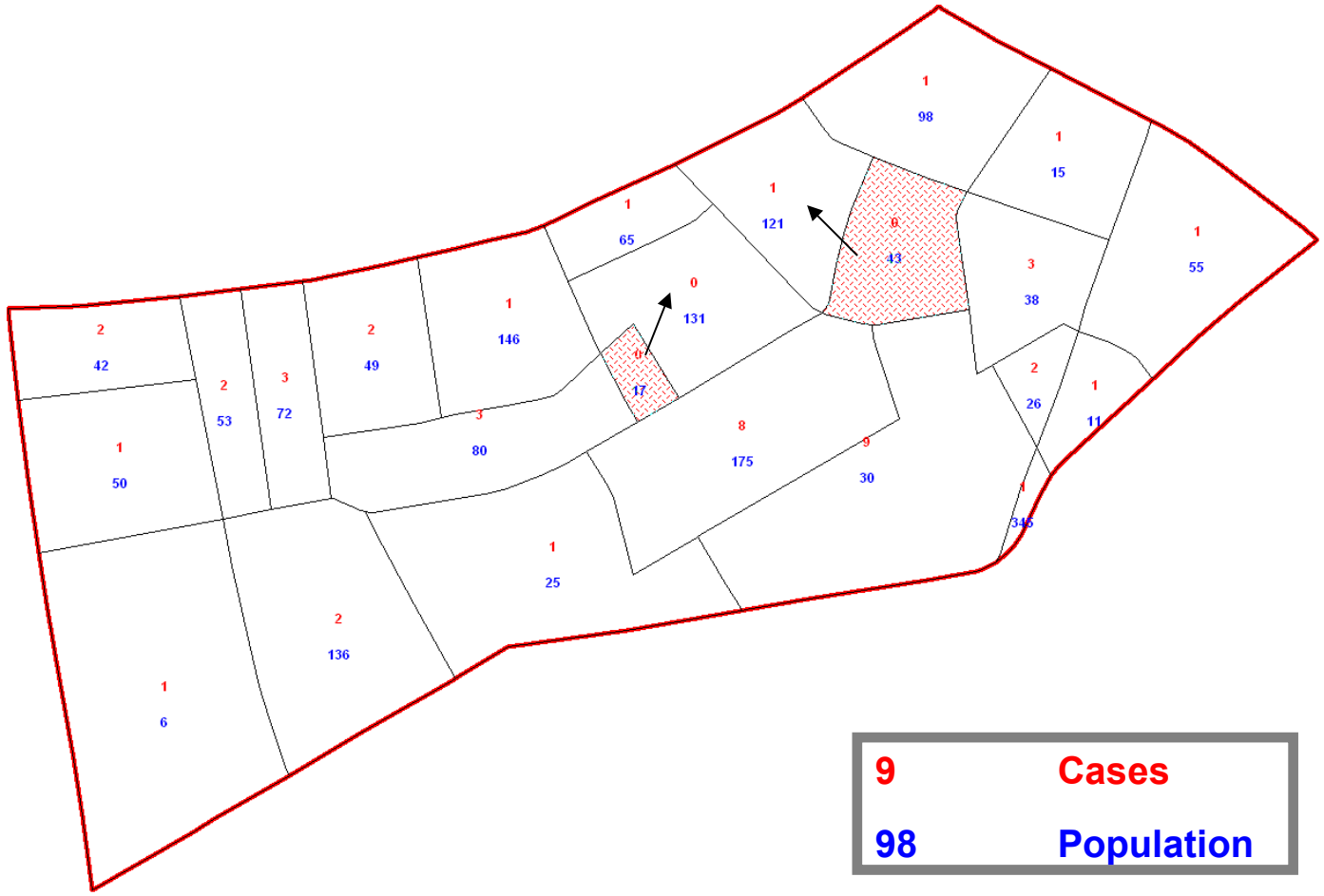


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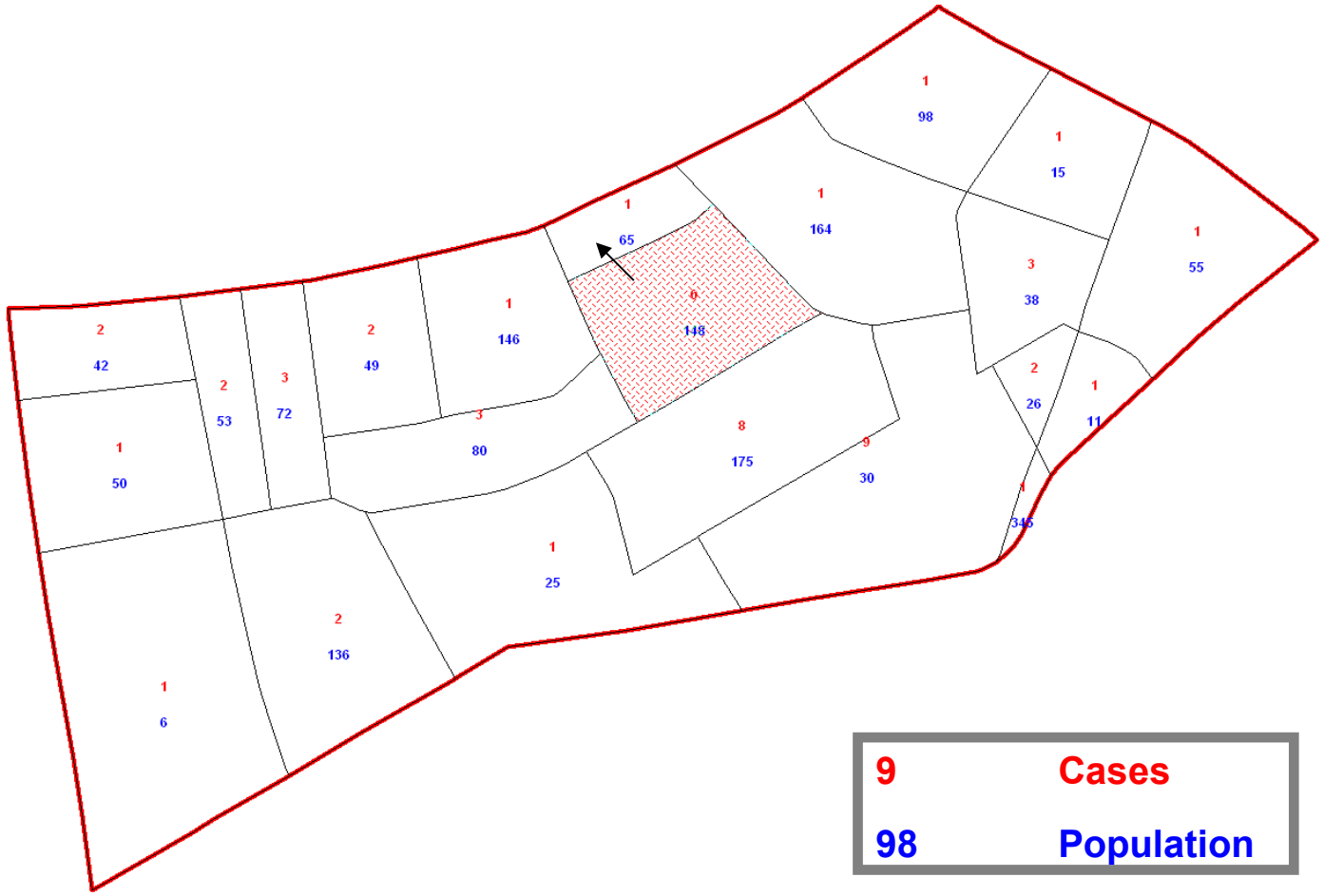


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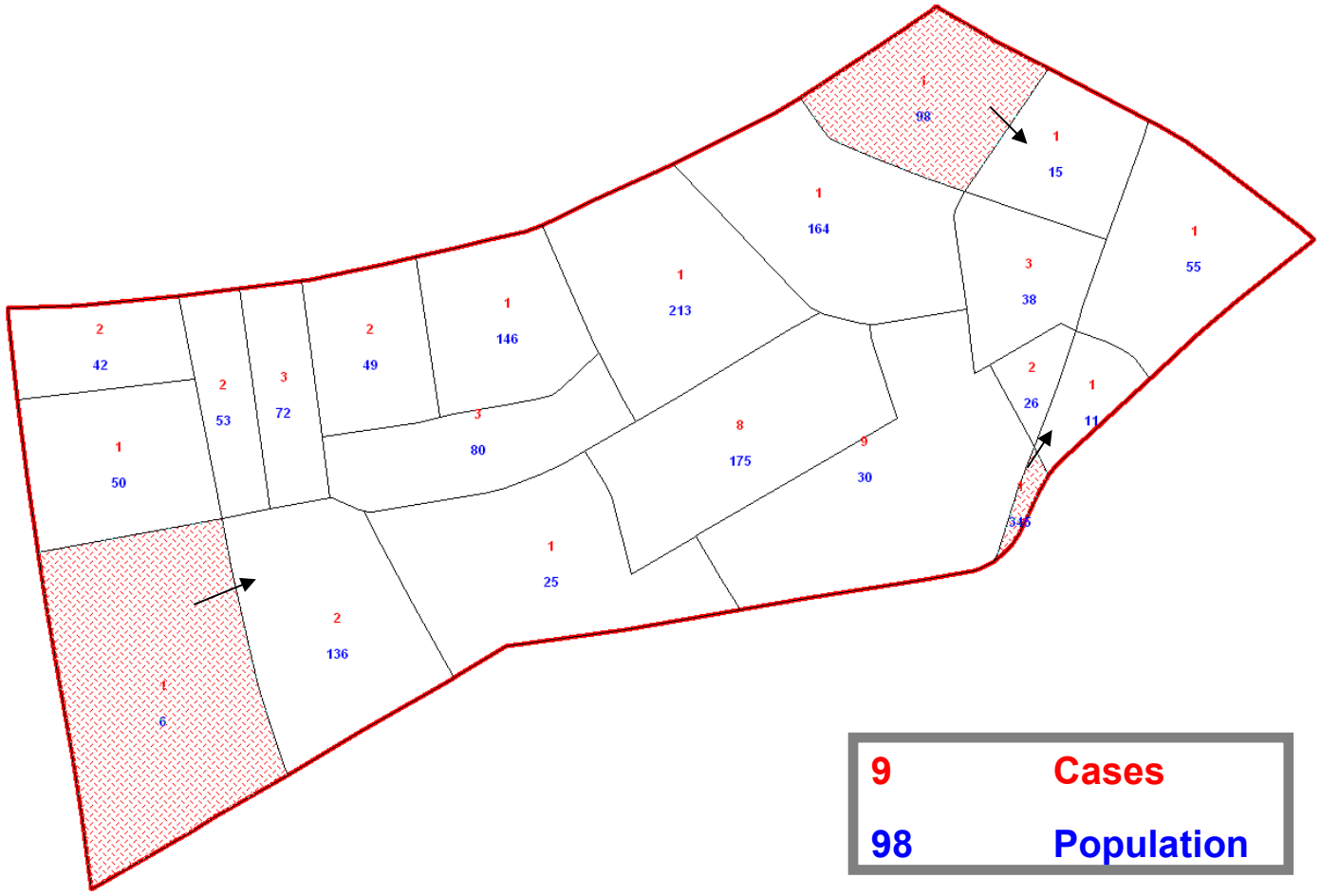


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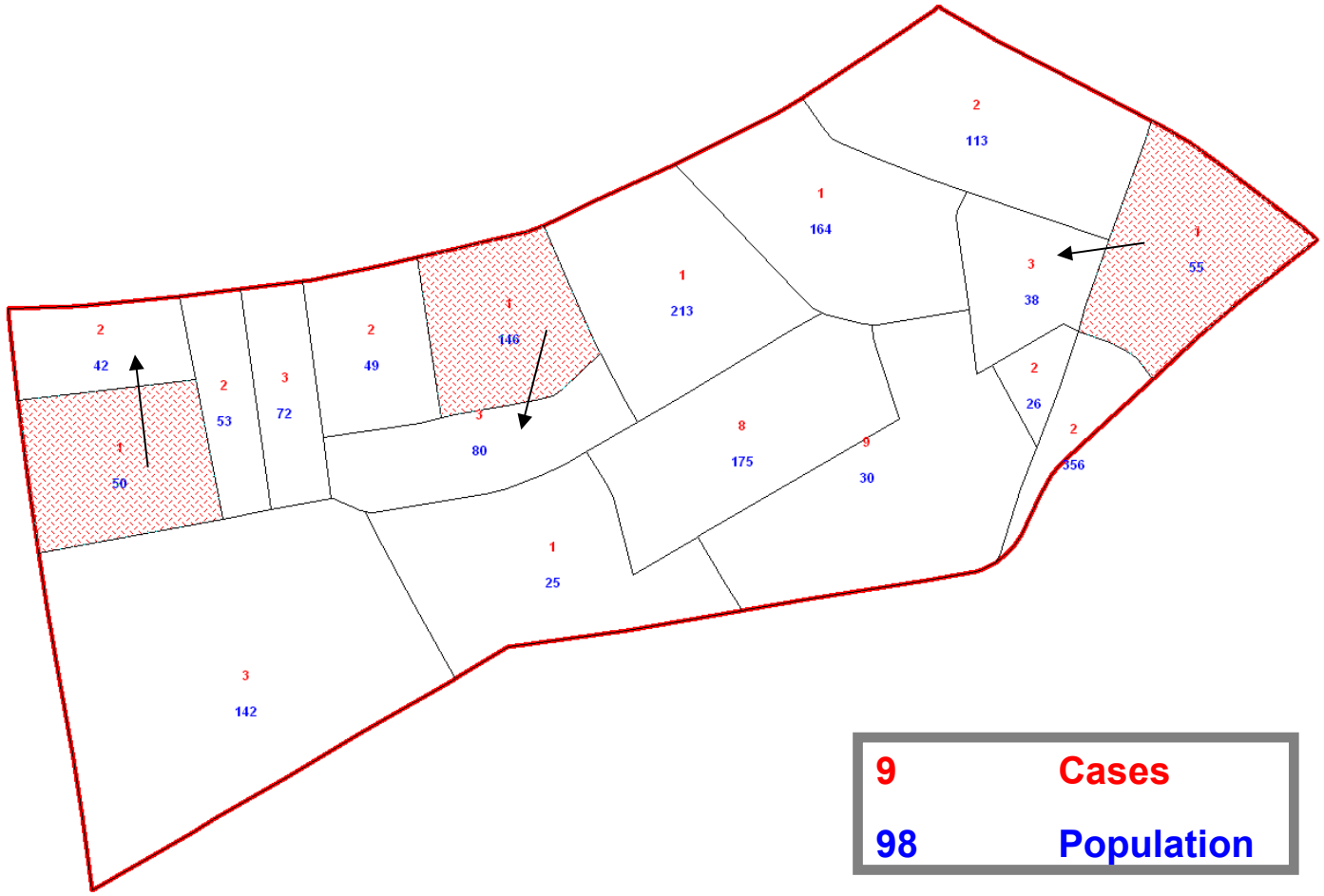


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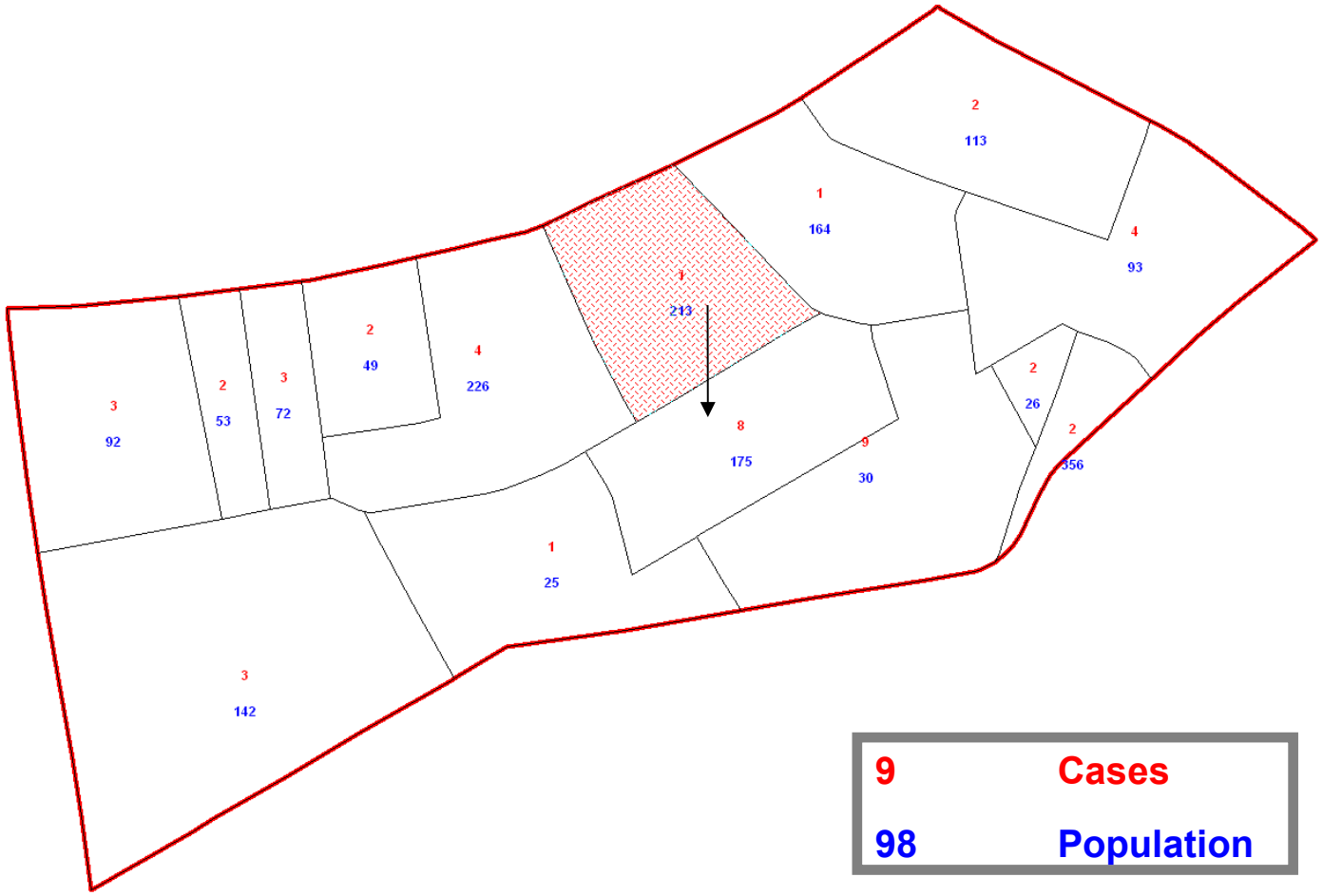


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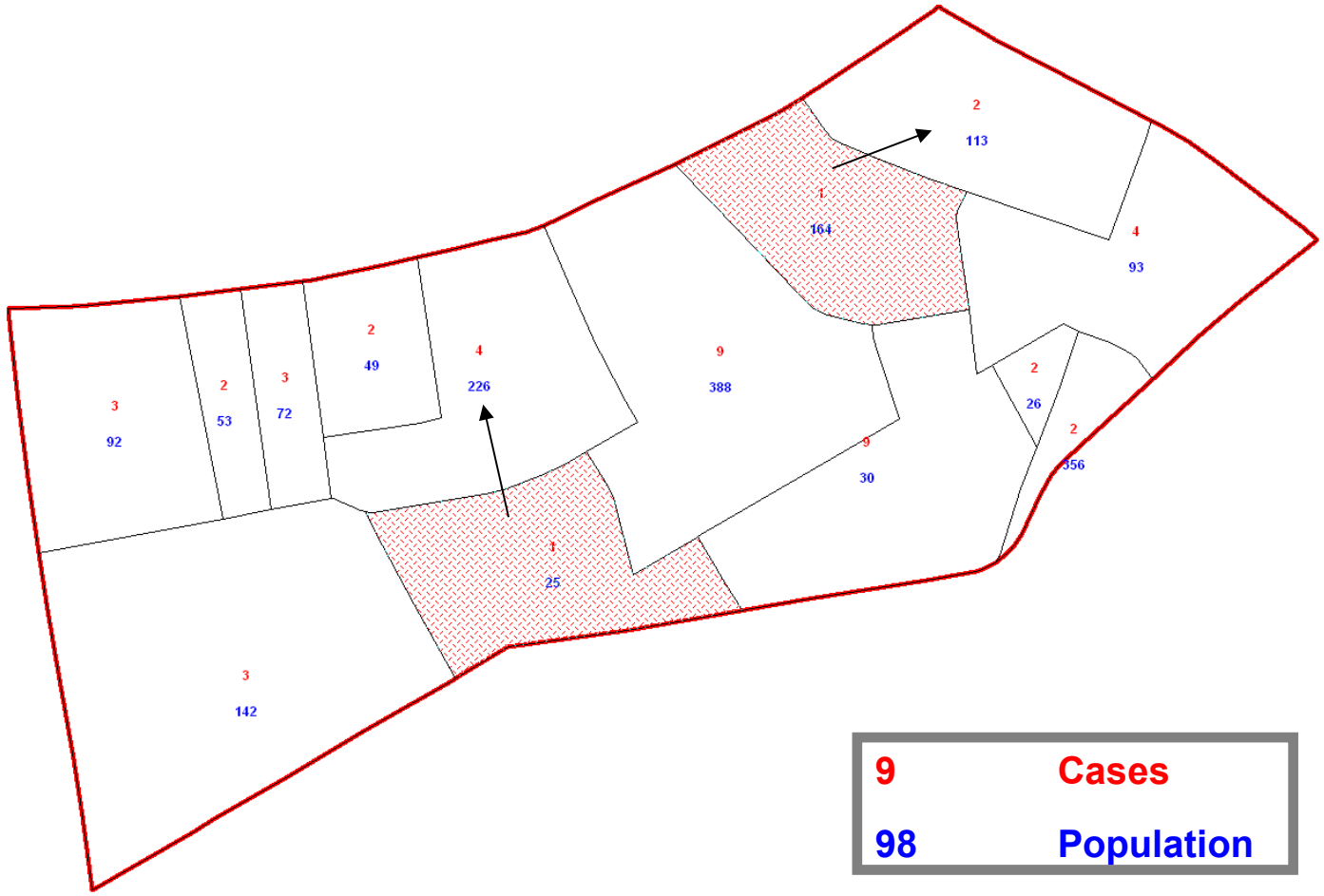


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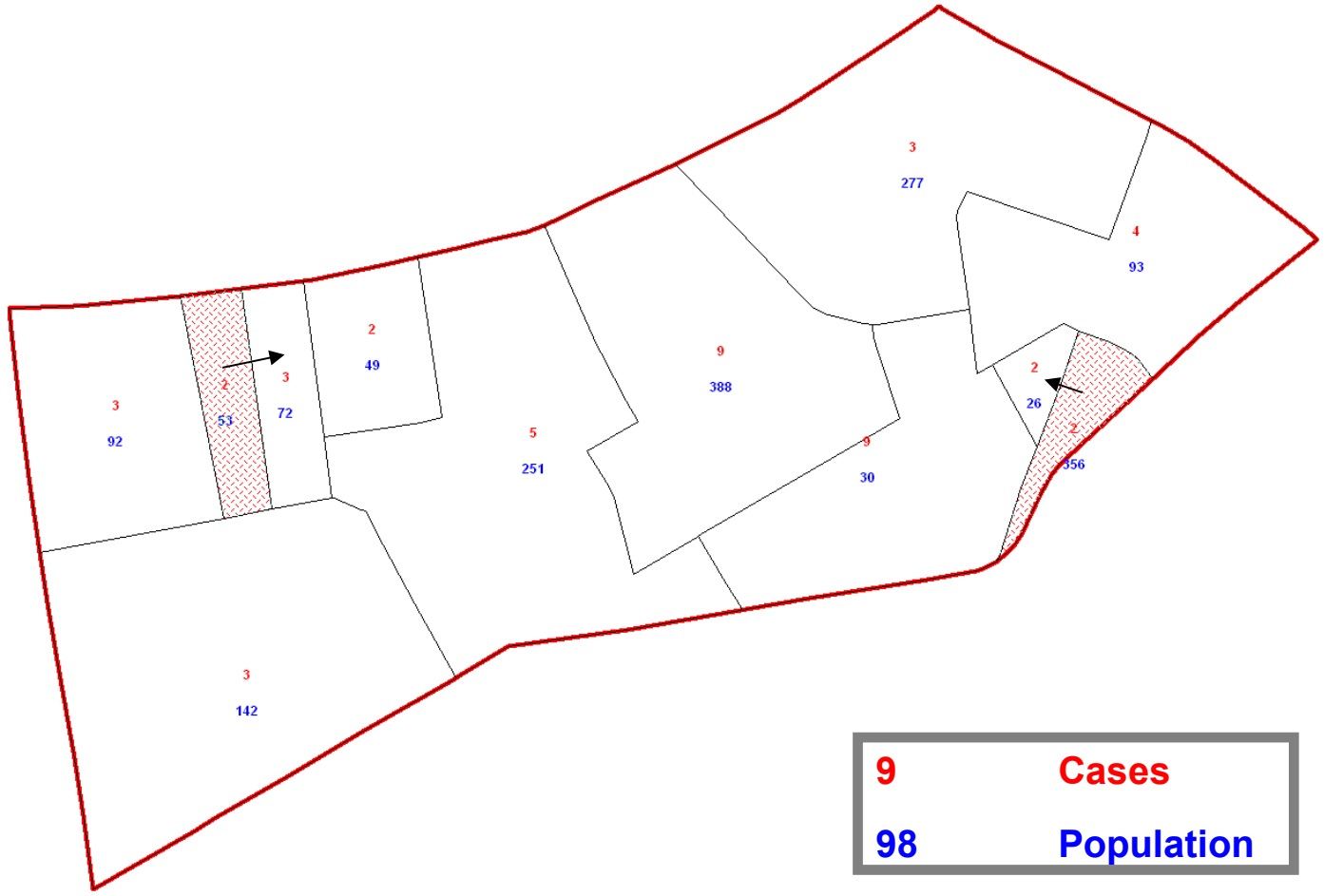


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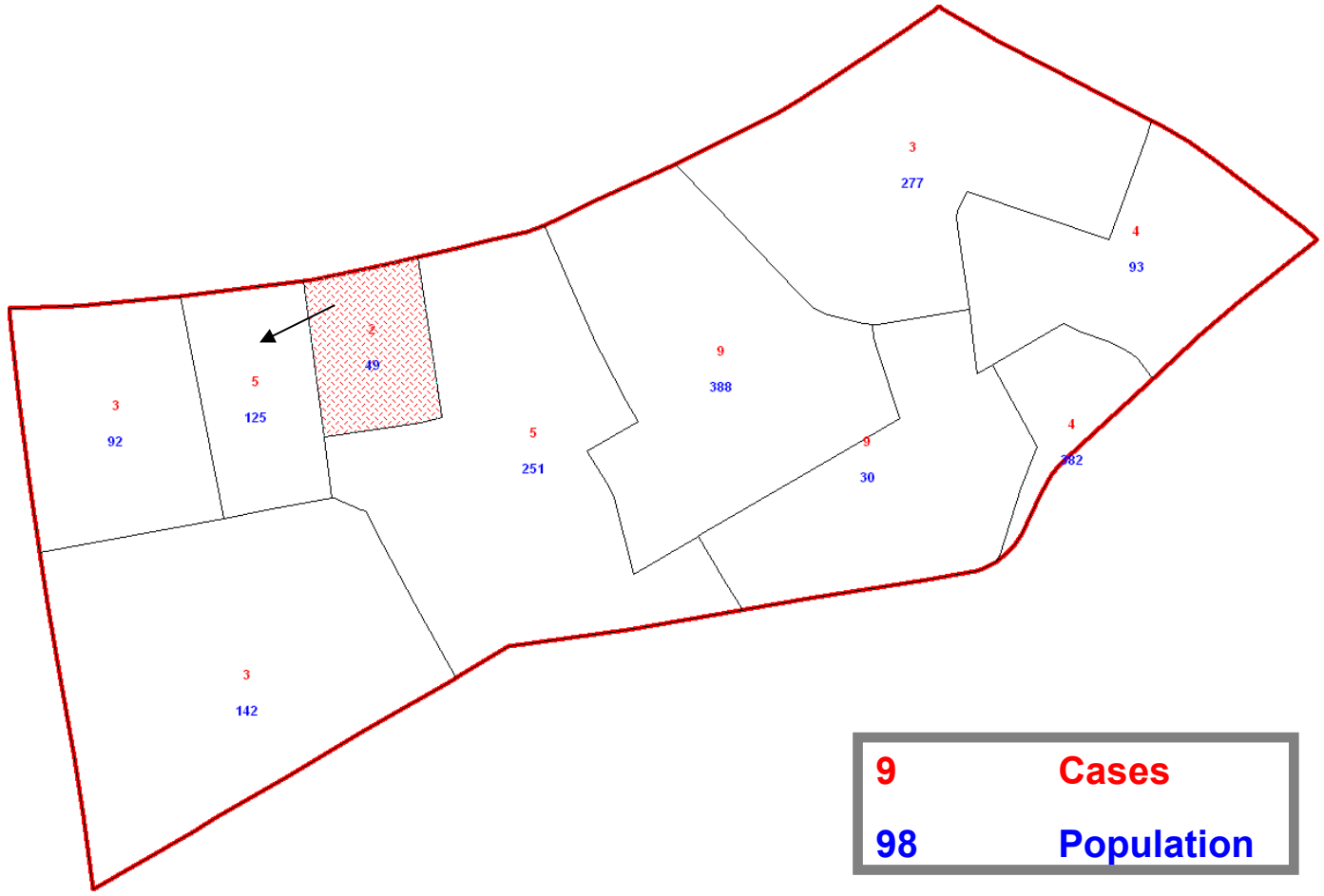


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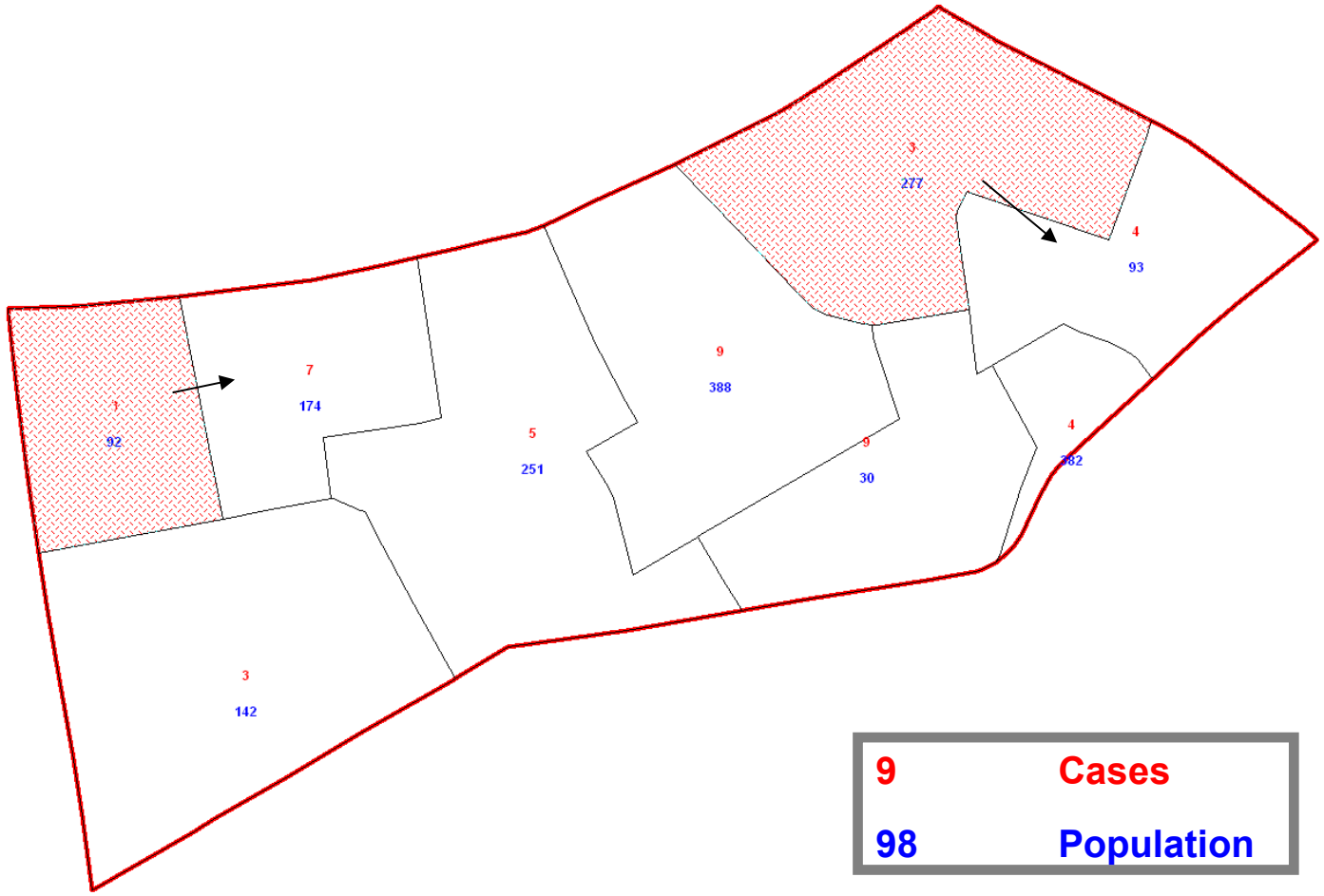


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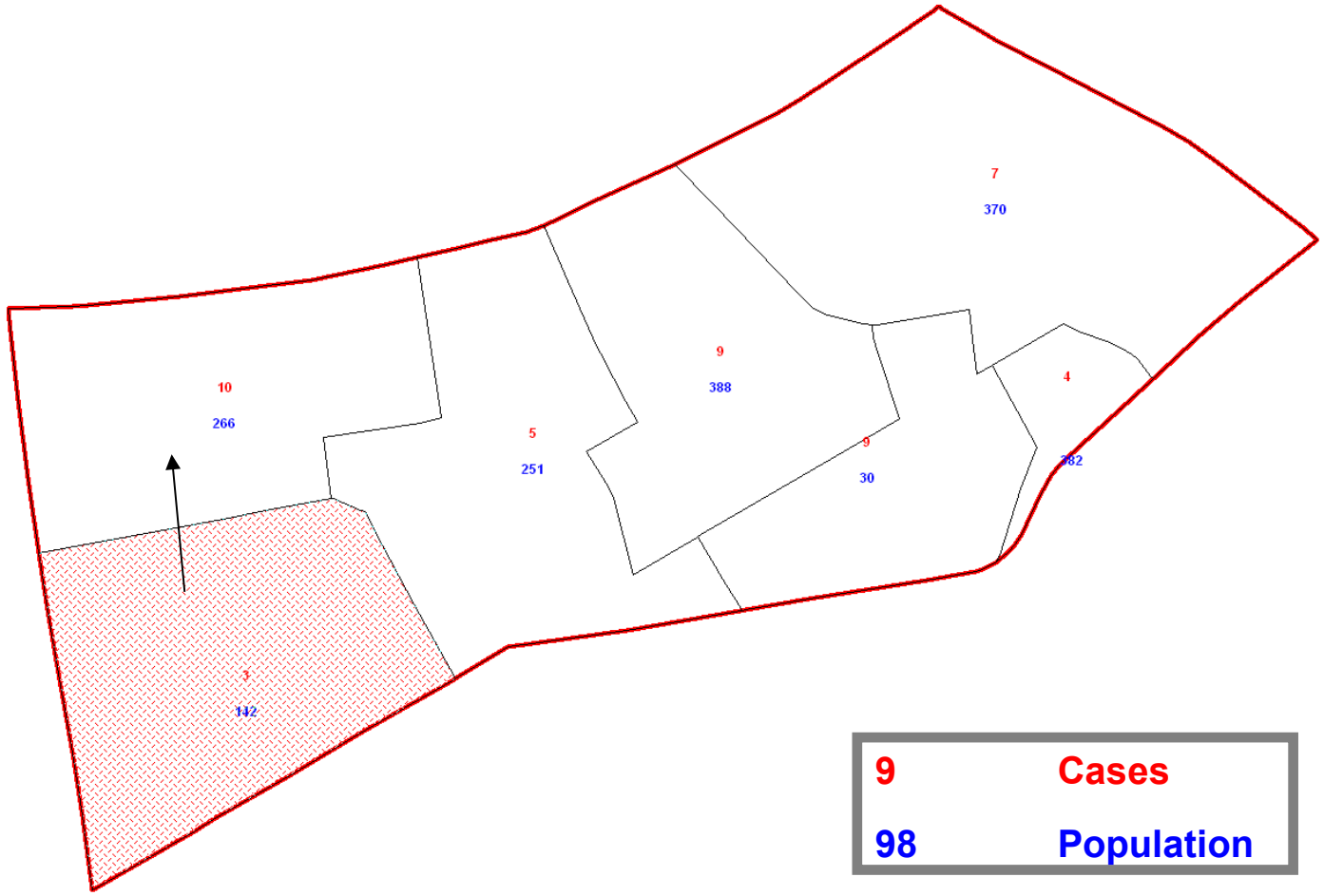


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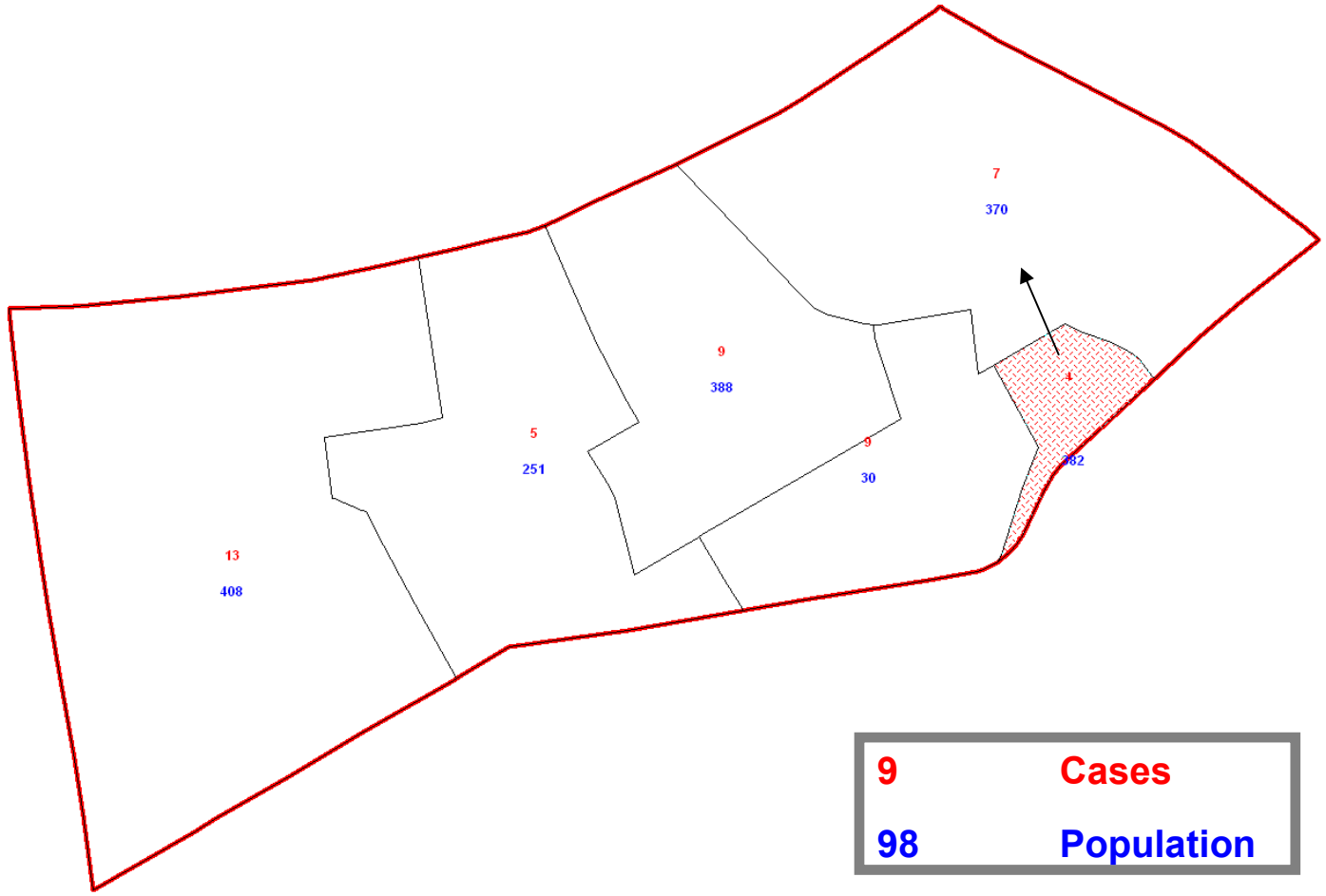


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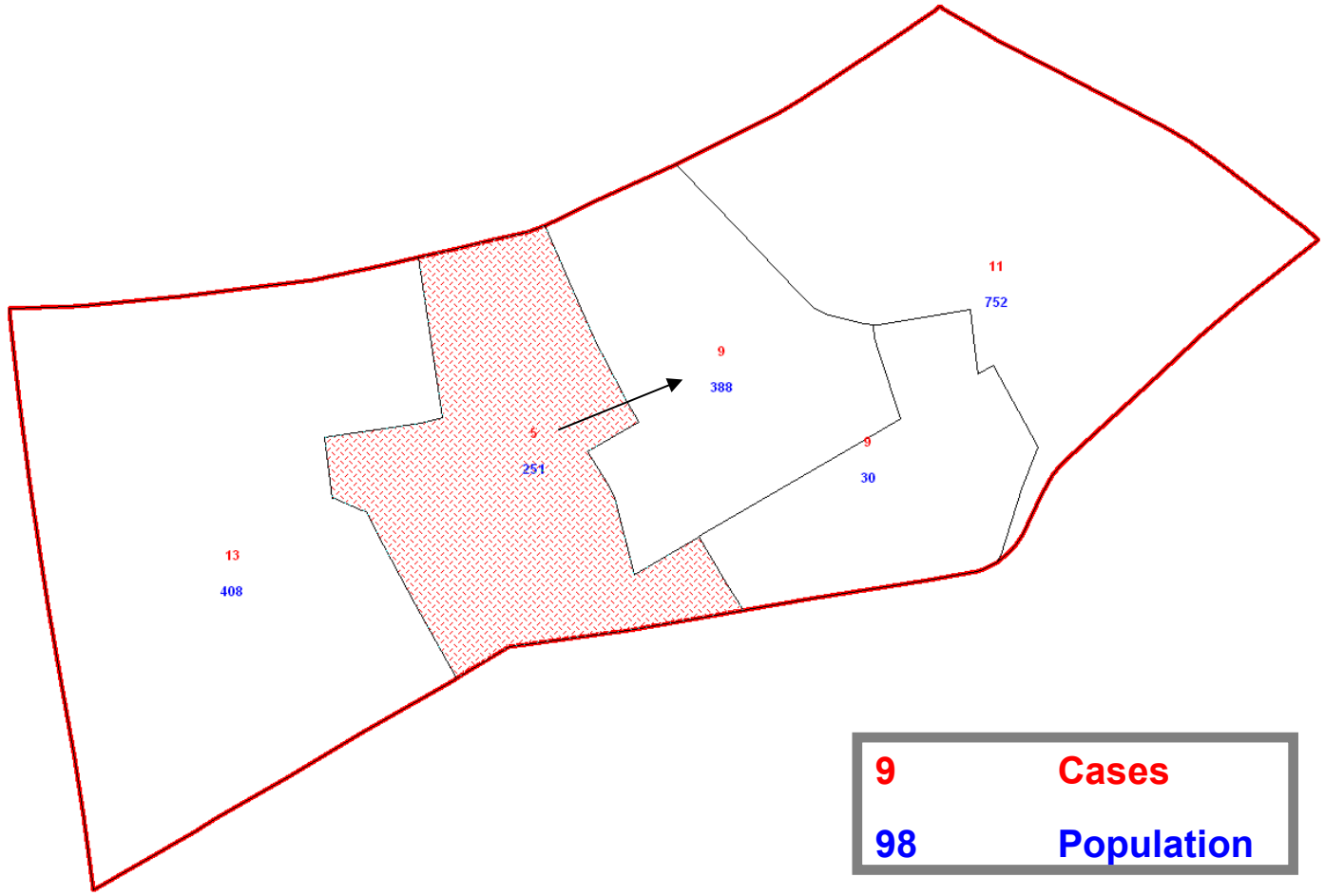


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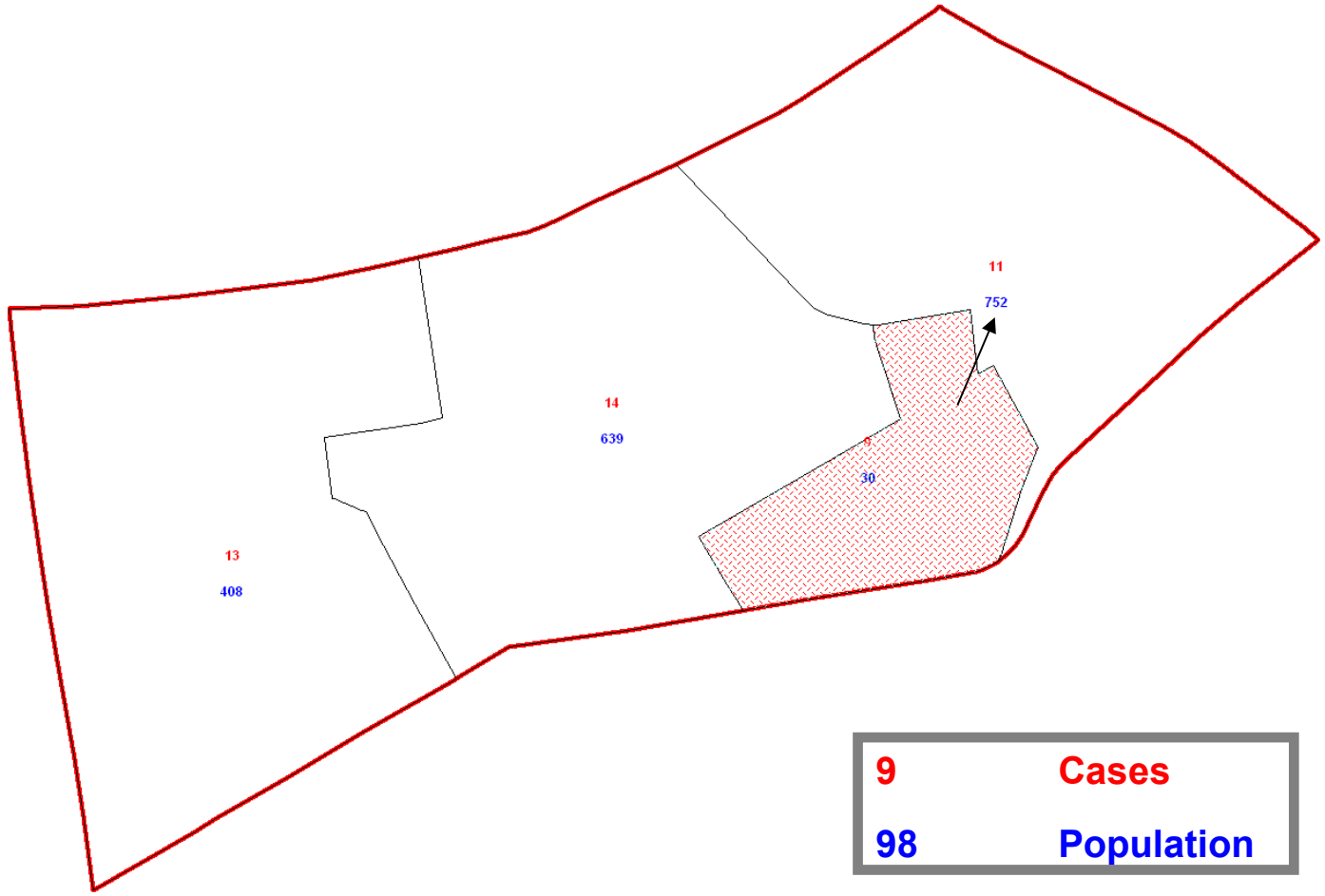


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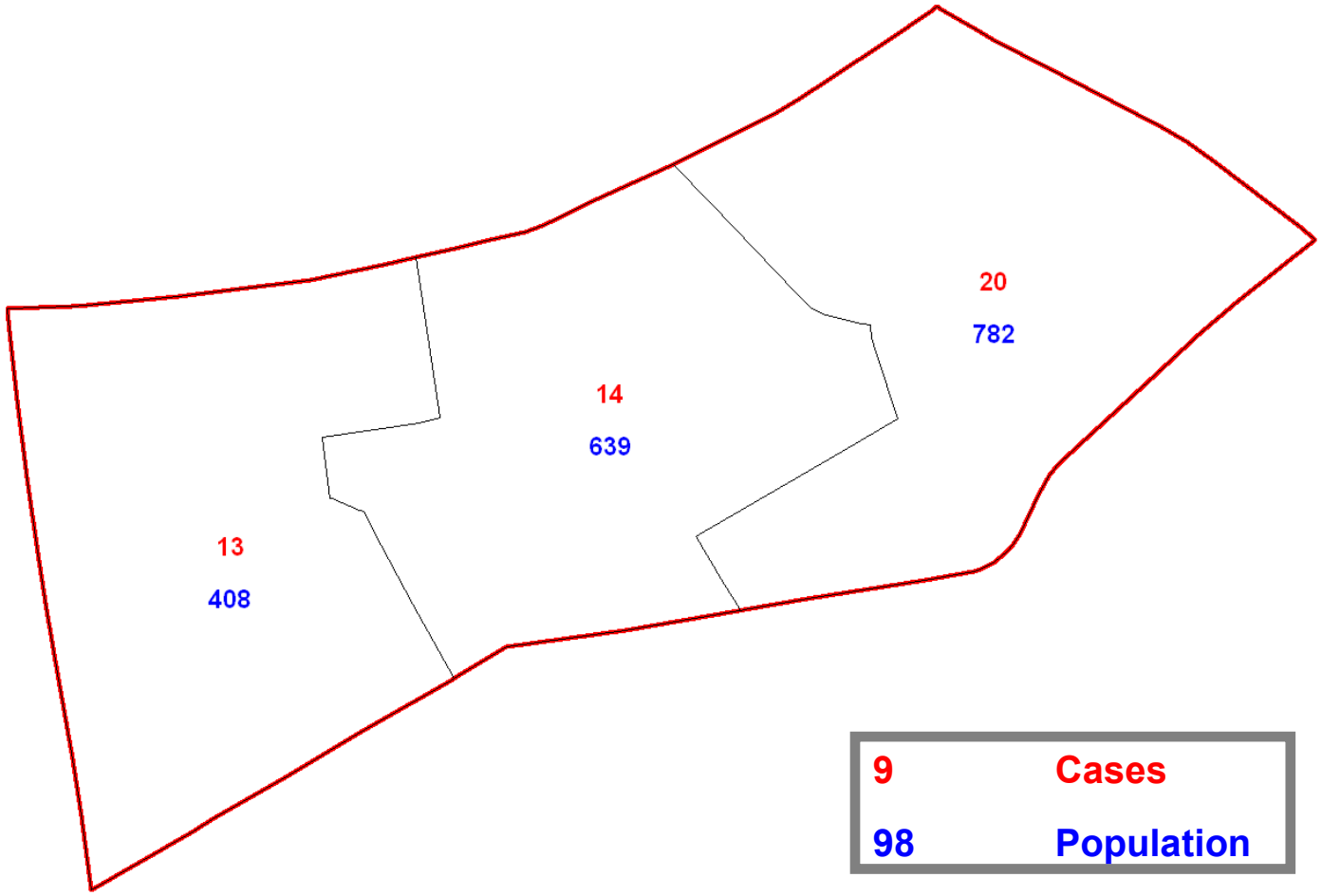


9	Cases
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The Aggregation Process: Example



9	Cases
98	Population

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Real Example: EPHT Sub-County

Number of Areas and Median 2010 Census Population		
Level of Aggregation	# Areas	Median Population
None*	4,919	3,734
7.5k minimum population	1,286	10,694
15k minimum population	779	22,287

* 2010 Census Tracts

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Real Example: EPHT Sub-County

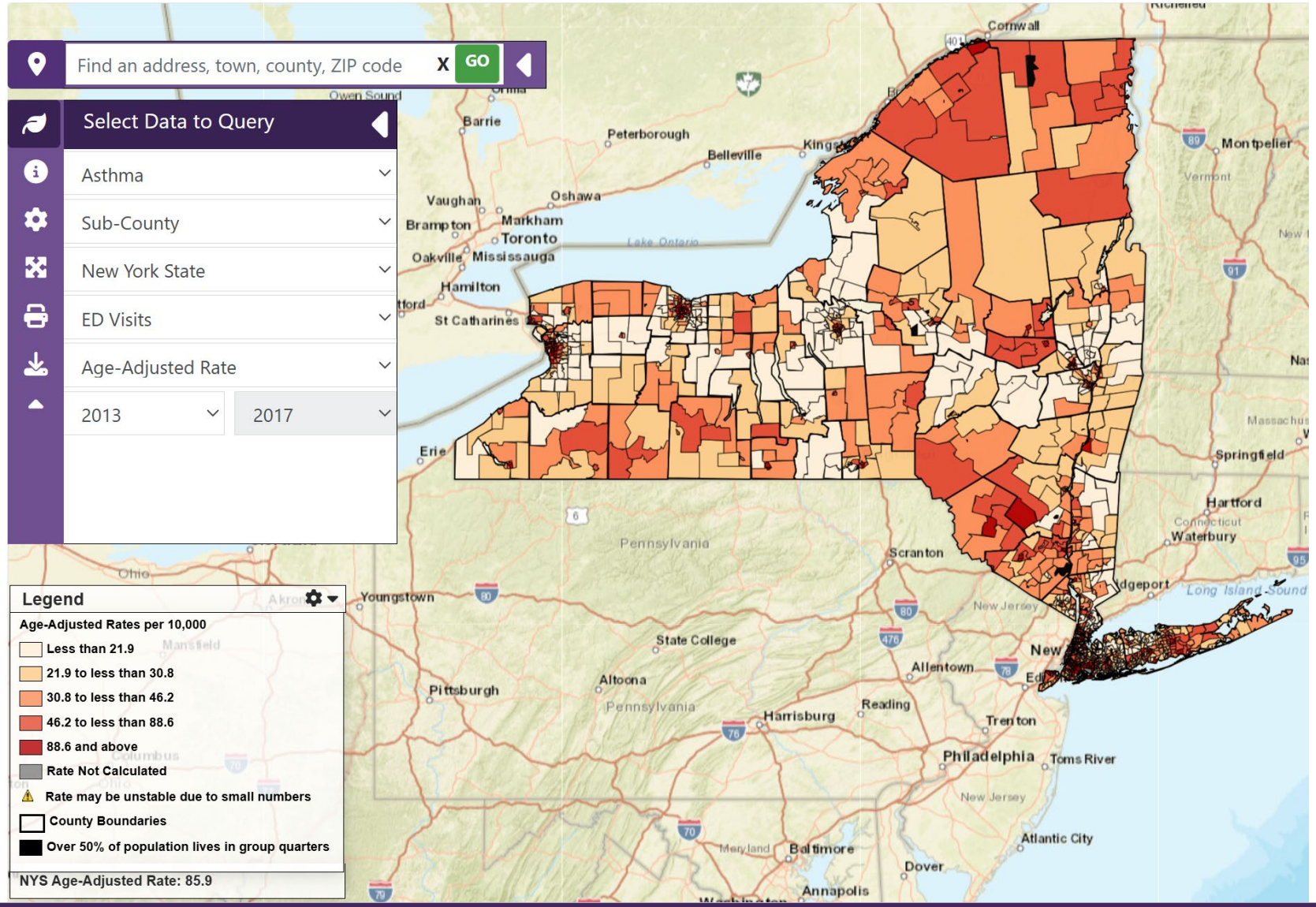
Counts and Percentage of 2010 Areas* Aggregated to 7.5K Minimum Population that have over 20 Cases (n=1,184)				
	2008-2012		2013-2017	
Indicator	**>20 cases**	**%**	**>20 cases**	**%**
Asthma ED	1184	100	1184	100
Asthma Hosp.	1123	94.85	1008	85.14
COPD ED	1183	99.92	1183	99.92
COPD Hosp.	1172	98.99	1176	99.32
Heart Attack Hosp.	1180	99.67	1180	99.67

* Excludes group quarters (n=53), zero-population (n=8), and parks (n=41)

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Real Example: EPHT Sub-County

Asthma Age-Adjusted ED Visits Rate per 10,000, New York State, 2013-2017



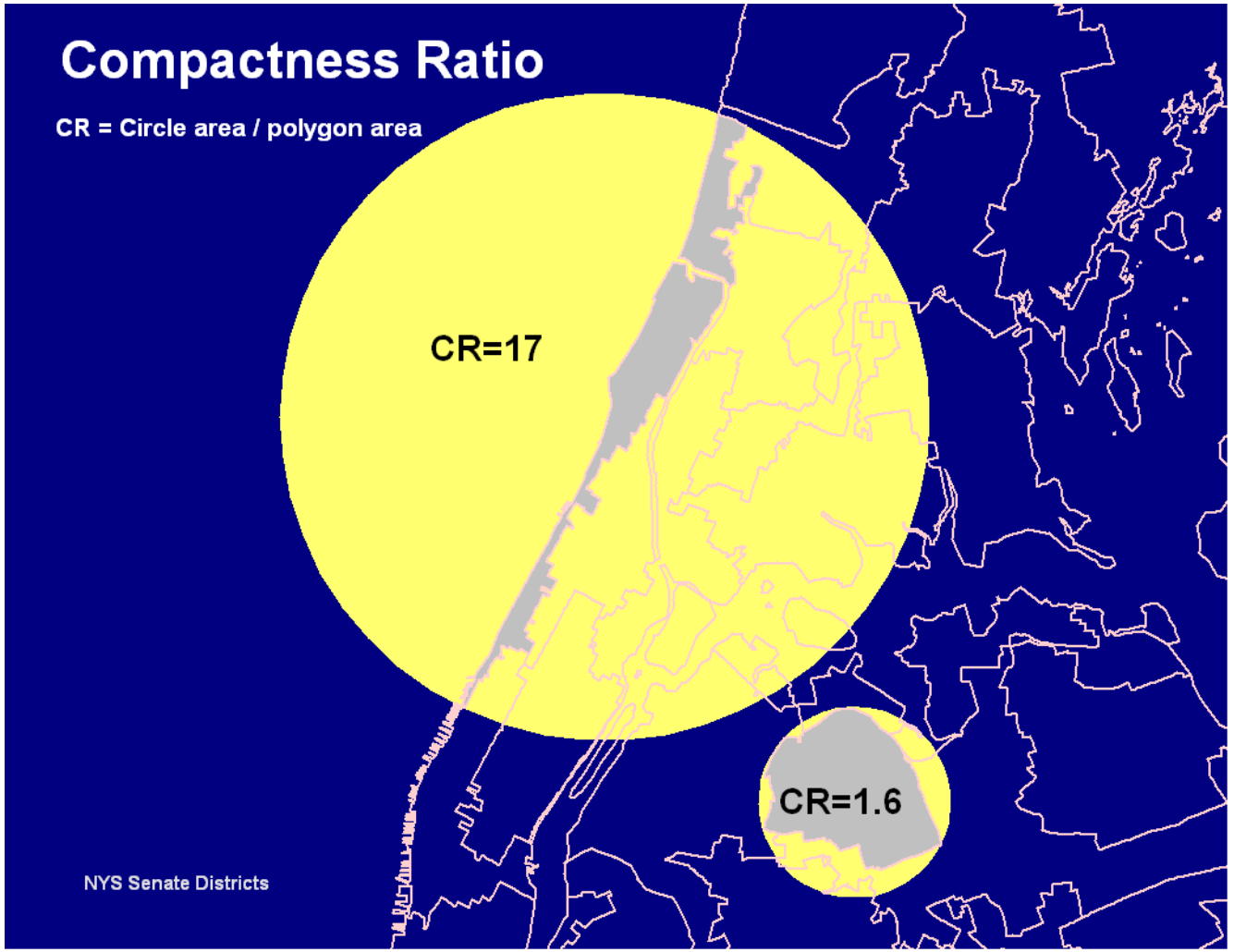
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Performance Measures

- Compactness
- Homogeneity with respect to demographic factors (measured as index of dissimilarity)
- Similar population sizes.
- Number of aggregated areas.
- Aggregated zones are completely contained within larger areas.
  - e.g. blocks aggregation areas contained within tracts

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Compactness Ratio



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Index of dissimilarity

The percentage of one group that would have to move to a different area in order to have an even distribution.

$$\frac{1}{2}\sum_{i=1}^{N}\left|\frac{b_i}{B}-\frac{w_i}{W}\right|$$

Wikipedia

b_i = the minority population of the ith area, e.g. census tract

B = the total minority population of the large geographic entity for which the index of dissimilarity is being calculated.

w_i = the non-minority population of the ith area

W = the total non-minority population of the large geographic entity for which the index of dissimilarity is being calculated.

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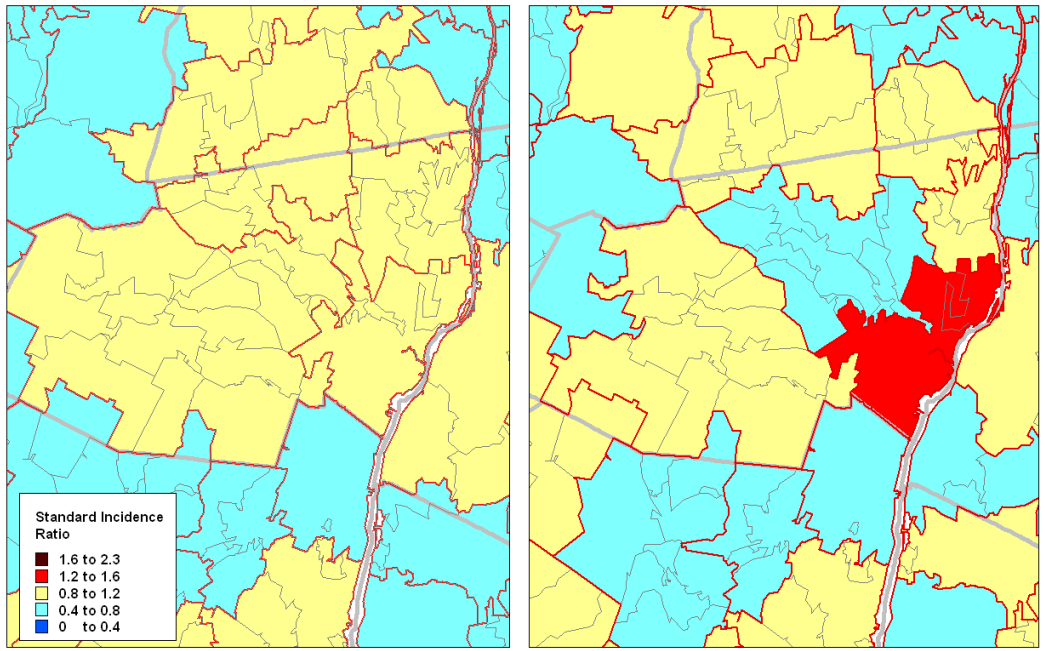
Follow-up Issues

Scale and method of aggregation will impact map & correlation coefficients.

Modifiable area unit problem

Aggregation

With Restrictions | **Without Restrictions**



Simulated Low Birth Weight SIRs. Each area 300 births

Aggregation Areas

Counties

ZIP Codes

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Some areas don't need to be smoothed!



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GAT Log File Contents

- NYSDOH Geographic Aggregation Tool log: version and date
- The input file and its characteristics
- The identification variable
- The boundary variable
- Output Shapefile and metadata
- Merge type and variable used, population file if used
- Any exclusion criteria
- Pre and post aggregation distributions
- Rate calculation information
- Additional information about outputs
- Warnings

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GAT Settings File

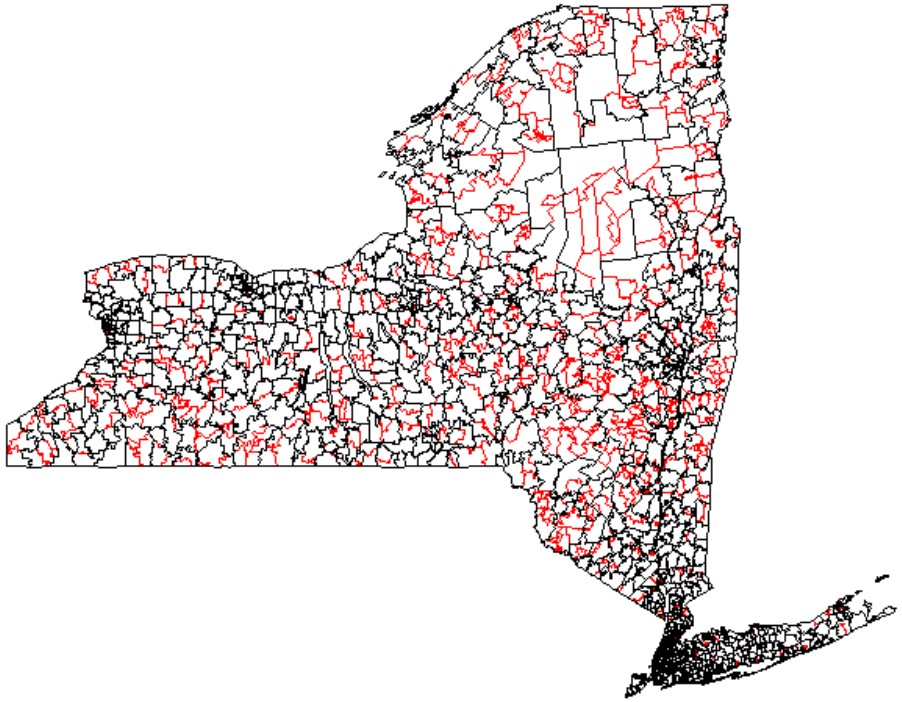
- *.Rdata file that can be used to replicate an aggregation
- runGATprogram(settings='path to Rdata file')



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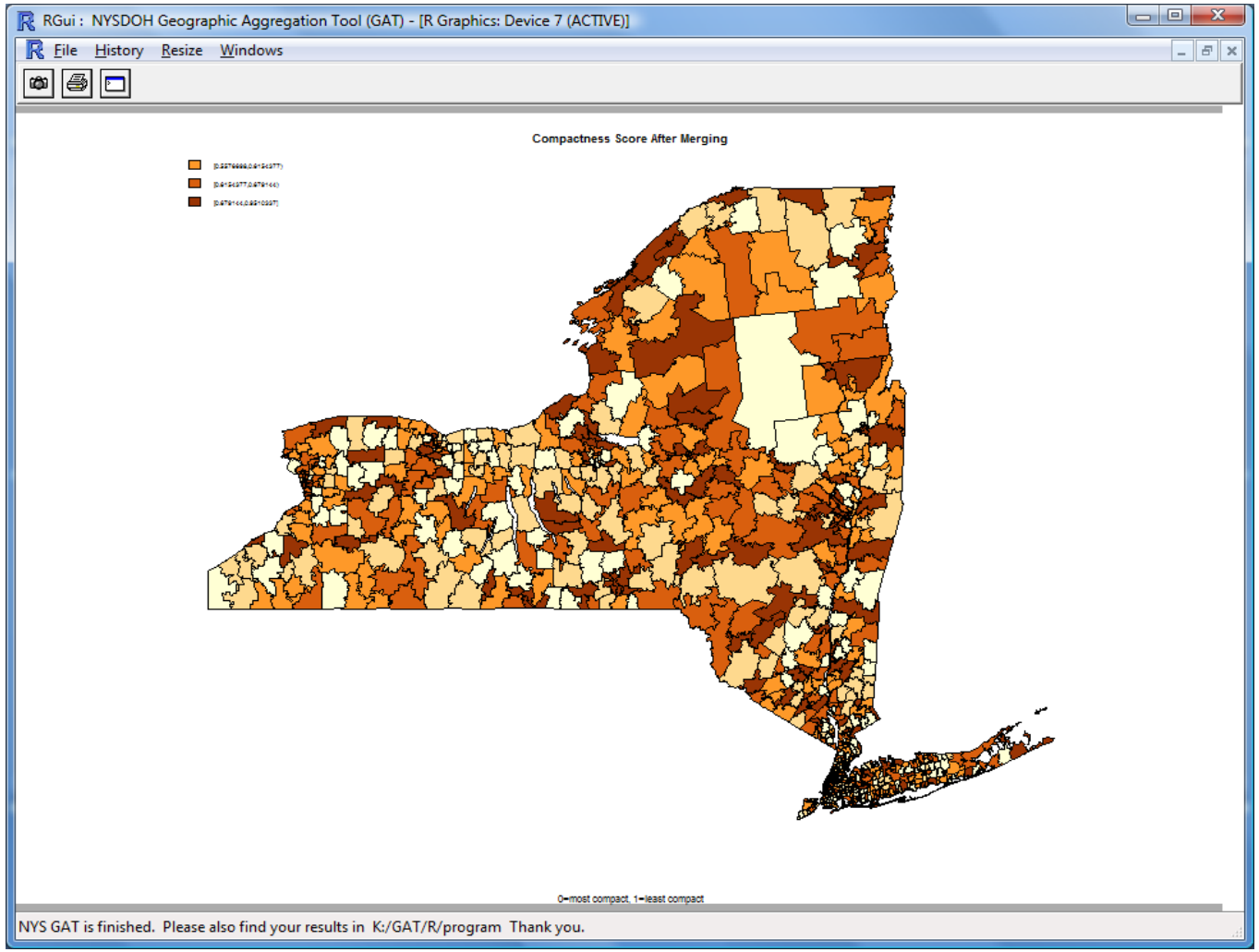
Compactness

Map of original areas (red) and aggregations (black)



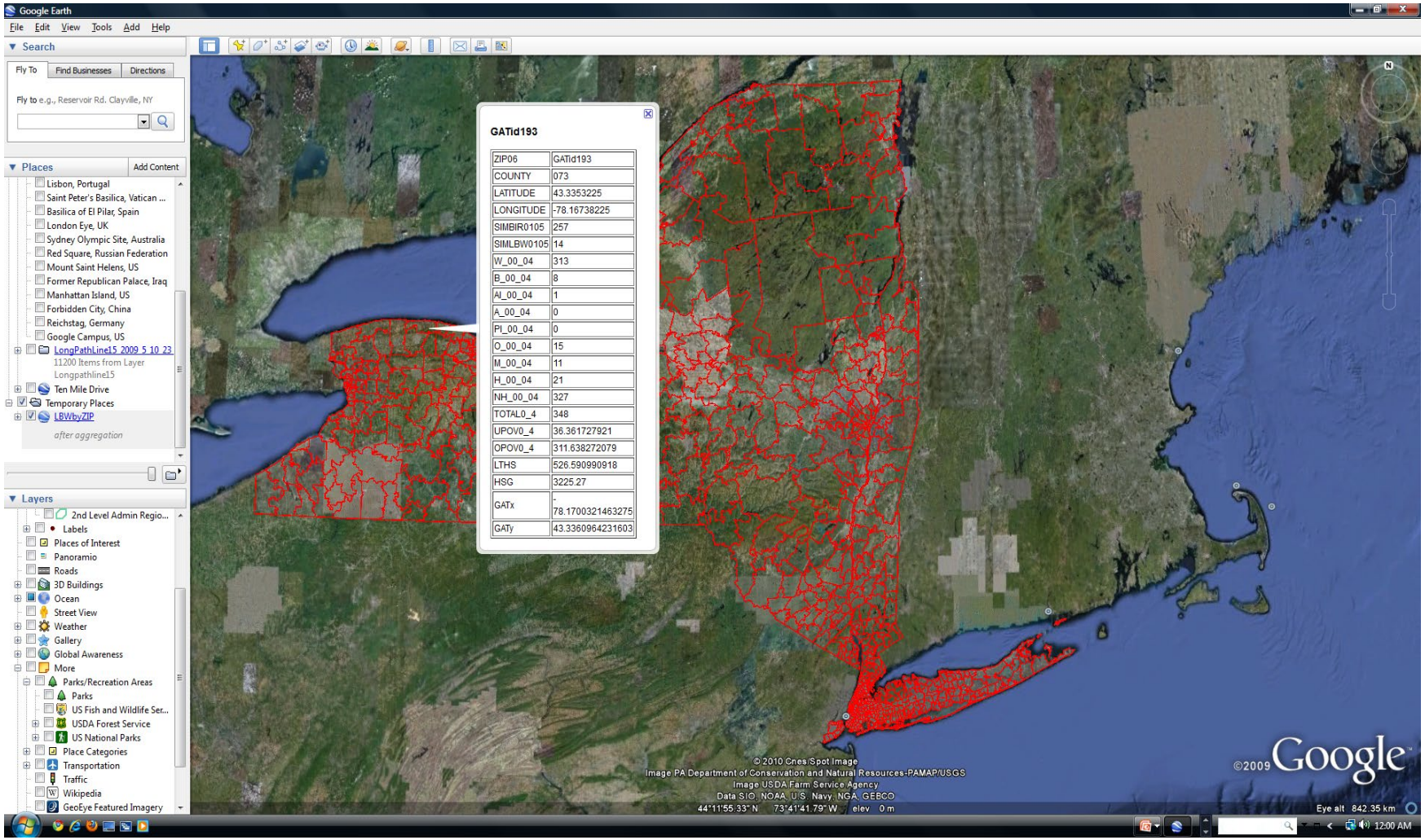
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Compactness



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GAT can output a KML Files



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The End